CUSTOMER CHURN PREDICTION

CAP STONE PROJECT

FINAL REPORT

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INTRODUCTION

PROBLEM UNDERSTANDING

a)Defining Problem Statement:

An E Commerce company provider is facing a lot of competition in the current market and it has become a challenge to retain the existing customers in the current situation. Hence, the company wants to develop a model through which they can do churn prediction of the accounts and provide segmented offers to the potential churners. In this company, account churn is a major thing because 1 account can have multiple customers. hence by losing one account the company might be losing more than one customer. You have been assigned to develop a churn prediction model for this company and provide business recommendations on the campaign.

OBJECTIVE: The objective is to develop a churn prediction model for this company and provide business recommendations on the campaign. The campaign suggestion should be unique and be very clear on the campaign offer because the recommendation will go through the revenue assurance team. If they find that you are giving a lot of free (or subsidized) stuff thereby making a loss to the company; they are not going to approve the recommendation. Hence there is a need to be careful while providing campaign recommendation.

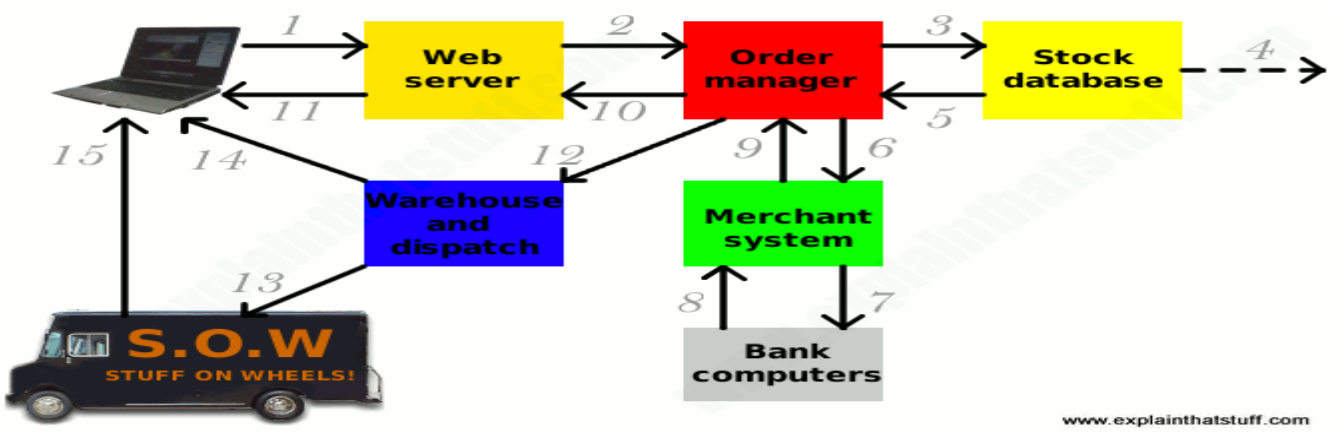
B) Need of the study/project :

Customer churn is what happens when a relationship of a customer with a company comes to the end. Customer churn rate is a rate at which a business is losing its clients. And while for subscription business a high customer churn can be very tedious, for e-commerce business model it is more typical to think about relationship with a client in terms of retention, rather than churn. It is easy to see how customers leave in subscription business (they simply press "Unsubscribe"), but how to know that customer is churned in e-commerce is the challenge

Customer churn prediction is highly important in e-commerce. The information about potential churners is valuable, because it allows you to take action when you still have time, and stop your customers from leaving. It is an essential action not only because you can lose additional revenue with every churn customer, but also because the buyer's spending before the churn date might not cover the initial spending on acquiring this buyer. It's a matter of a fact, that retaining an existing buyer is cheaper than getting a new one. It costs five time as much to attract a new customer than to keep an existing one.

Hence by digging deep and finding the insights and providing the recommendations we can benefit from the project interms of learning things about prediction , how the business model works, what it takes to understand the churn mechanism, monetary benefits and so on…

c) Understanding business/social opportunity:



The simple definition of eCommerce is that it is a business model that allows people to buy and sell products and services on the internet. Nowadays, you can buy almost anything through your computer, tablet, or mobile device. It's this convenience that's making it one of the fastest-growing markets.Almost every retail business now has an eCommerce shop as an addition to their brick-and-mortar ones. Online-only retail business numbers are growing each day, too, and many shops have closed their physical doors in favor of an online storefront. After all, sellers want to be where their customers are.

An eCommerce system doesn't differ much from a regular physical store. It all starts with a product you want to sell. From there, you would need a place to sell the product. In eCommerce, your website is your store - the site takes on the role of the shelves and display, and your customers use it to browse and purchase your goods.The internet is a crowded place, so it will be hard for your shoppers to just stumble upon your store. You have to attract them, so catchy and carefully planned eCommerce marketing will be crucial to your success.

Once customers start shopping on your site, they'll have to be able to order and pay you for the products. You'll also need to put some security measures in place, as you'll be dealing with a lot of sensitive information.
Acquiring a new customer can cost five times more than retaining an existing customer. Increasing customer retention by 5% can increase profits from 25-95%.As customer churn directly impacts both the top-line and bottom-line revenue of the business, existing customer base needs to be protected. Providing all customers with offers to retain them would make a dent in the profitability and hence it is very important to focus only on select set of customers who are at a higher risk of churning.

1) This project aims at identifying customers who have a propensity to churn so that targeted campaign offers can be provided to them. This would ensure better top-line and bottom-line revenue for the business.
2) Other insights from Exploratory data analysis and best performing model(s) will also be used to come up with business recommendations.

2)DATA REPORTING

Understanding how data was collected in terms of time, frequency and methodology:

• The dataset contains 'Account' level data which is master data along with features of account such as gender, marital status, city tier, account user count of primary account holder, whether the account is live or churned, segment the account belongs to. It also possibly contains certain derived features such as tenure (which probably could have been derived from account open date)
• The dataset also contains information taken/derived from transaction data and rolled up at Account level and given as a feature for the account – for e.g., Number of days since none of the account holders contacted customer care, monthly average cashback for last 12 months, number of complaints made last year, number of times customer care contacted last year, revenue per month in last 12 months, how many times customers have used coupons to pay in last 12 months, satisfaction score and customer service score.
• Most of the transaction data roll-up at account level has been done for 12 months (previous year). However, the revenue growth percentage has been taken for last year in comparison with previous one year which implies that 24-months' worth of data has been used to calculate this field.
• As the dataset has been provided, the methodology used by customer to extract data is not known. The frequency at which this dataset is extracted has also not been specified.

Visual inspection of data (rows, columns, descriptive details):

- The dataset has 11260 rows and 19 columns.
- There are 5 columns of float type, 2 columns of integer type and 12 columns of object type.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11260 entries, 0 to 11259
Data columns (total 19 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   AccountID                11260 non-null  int64
 1   Churn                    11260 non-null  int64
 2   Tenure                   11158 non-null  object
 3   City_Tier                11148 non-null  float64
 4   CC_Contacted_LY          11158 non-null  float64
 5   Payment                  11151 non-null  object
 6   Gender                   11152 non-null  object
 7   Service_Score            11162 non-null  float64
 8   Account_user_count       11148 non-null  object
 9   account_segment          11163 non-null  object
 10  CC_Agent_Score           11144 non-null  float64
 11  Marital_Status           11048 non-null  object
 12  rev_per_month            11158 non-null  object
 13  Complain_ly              10903 non-null  float64
 14  rev_growth_yoy           11260 non-null  object
 15  coupon_used_for_payment  11260 non-null  object
 16  Day_Since_CC_connect     10903 non-null  object
 17  cashback                 10789 non-null  object
 18  Login_device             11039 non-null  object
dtypes: float64(5), int64(2), object(12)
memory usage: 1.6+ MB
```

There are several columns that are supposed to be read as numeric, instead they have been read as object type for e.g., Tenure is a numeric field but has been read as object. Those columns need to be checked for special characters and need to be treated before the column can be changed to numeric for further processing.

• There are no duplicate rows in the data set. Each account id has one unique row.

• Several columns have null values

• The following table shows number of rows containing nulls and special characters that require data cleaning. All special characters present in the data set were treated with nulls so that they can be imputed. Some columns such as Gender and account_segment contained multiple values to represent the same category for e.g., 'M', 'Male'. Cleaning up of those values was also performed.

Below is the Descriptive statistical report of the dataset:

33]:

	AccountID	Churn	City_Tier	CC_Contacted_LY	Service_Score	CC_Agent_Score	Complain_ly
count	11260.00000	11260.000000	11148.000000	11158.000000	11162.000000	11144.000000	10903.000000
mean	25629.50000	0.168384	1.653929	17.867091	2.902526	3.066493	0.285334
std	3250.62635	0.374223	0.915015	8.853269	0.725584	1.379772	0.451594
min	20000.00000	0.000000	1.000000	4.000000	0.000000	1.000000	0.000000
25%	22814.75000	0.000000	1.000000	11.000000	2.000000	2.000000	0.000000
50%	25629.50000	0.000000	1.000000	16.000000	3.000000	3.000000	0.000000
75%	28444.25000	0.000000	3.000000	23.000000	3.000000	4.000000	1.000000
max	31259.00000	1.000000	3.000000	132.000000	5.000000	5.000000	1.000000

By looking at the above table we can conclude the following:

The following table shows a basic statistical description of the numeric columns after data clean-up. It contains a 5-point summary of the numeric fields – minimum, maximum, 25th percentile, 50th percentile and 75th percentile. In addition, it contains count of values present in each column, mean and standard deviation.Tenure seems to have a very huge range up to 99. It could be in months in which case those would be valid values.

• The maximum limit for many of the variables seems to be very far apart from the 75th percentile for many variables such as cashback, revenue per month and customer contacted last year. There seems to be significant positive skew in these variables. A look at the boxplot and histogram will confirm the presence of outliers.

Understanding of attributes (variable info, renaming if required):

Data Dictionary: (About the data)

AccountID: Account unique identifier

Churn: Account churn flag-(Leaving company)(Y/N) - Target variable

Tenure: How long customer is associated with the company

City_Tier: City which customers belong

CC_Contacted_L1 2m: how many times all the customers of the account has contacted customer care in last 12months

Payment: Preferred Payment mode of the customers in the account

Gender: Gender of the primary customer of the account

Service_Score: Satisfaction score given by customers of the account on service provided by company

Account_user_count: How many customers have been associated to the particular account

account_segment: Account segmentation on the basis of spend

CC_Agent_Score: Satisfaction score given by customers of the account on customer care service provided by company

Marital_Status: Marital status of the primary customer of the account

rev_per_month: Monthly average revenue generated by account in last 12 months

Complain_l12m: Any complaints has been raised by account in last 12 months

rev_growth_yoy: revenue growth percentage of the account (last 12 months vs last 24 to 13 month)

coupon_used_l12 m: How many times customers have used coupons to do the payment in last 12 months

Day_Since_CC_connect: Number of days since no customers in the account has contacted customer care

cashback_l12m: Monthly average cashback generated by account in last 12 month

Login_device: Preferred login device of the customers in the account

Columns such as CC_Contacted_L1 2m , Complain_l12m, Day_Since_CC_connect, etc can be renamed to shorter names for accessing convenience. The variable names would be changed later to shorten or make it uniform when one hot encoding is done in a later section.

Look in the below table for clearer understanding on variable values along with nulls, data clean-up requirements etc..

Column	Values present	% Rows with values present	Number of Nulls	% Rows with nulls	Data clean-up needed?	% Rows needing data cleaning
AccountID	11260	100.00%	0	0%	None	0.00%
Churn	11260	100.00%	0	0%	None	0.00%
Tenure	11158	99.09%	102	0.91%	Yes - #	1.03%
City_Tier	11148	99.01%	112	0.99%	None	0.00%
CC_Contacted_LY	11158	99.09%	102	0.91%	None	0.00%

Payment	11151	99.03%	109	0.97%	None	0.00%
Gender	11152	99.04%	108	0.96%	Yes - M,F	5.74%
Service_Score	11162	99.13%	98	0.87%	None	0.00%
Account_user_count	11148	99.01%	112	0.99%	Yes - @	2.95%
account_segment	11163	99.14%	97	0.86%	Yes-Regular + Super +	2.74%
CC_Agent_Score	11144	98.97%	116	1.03%	None	0.00%
Marital_Status	11048	98.12%	212	1.88%	None	0.00%
rev_per_month	11158	99.09%	102	0.91%	Yes - +	6.12%
Complain_ly	10903	96.83%	357	3.17%	None	0.00%
rev_growth_yoy	11260	100.00%	0	0.00%	Yes - $	0.03%
coupon_used_for_payment	11260	100.00%	0	0.00%	Yes - $, *, #	0.03%
Day_Since_CC_connect	10903	96.83%	357	3.17%	Yes - $	0.01%
Cashback	10789	95.82%	471	4.18%	Yes - $	0.02%
Login_device	11039	98.04%	221	1.96%	Yes - &&&&	4.79%

EXPLORATORY DATA ANALYSIS

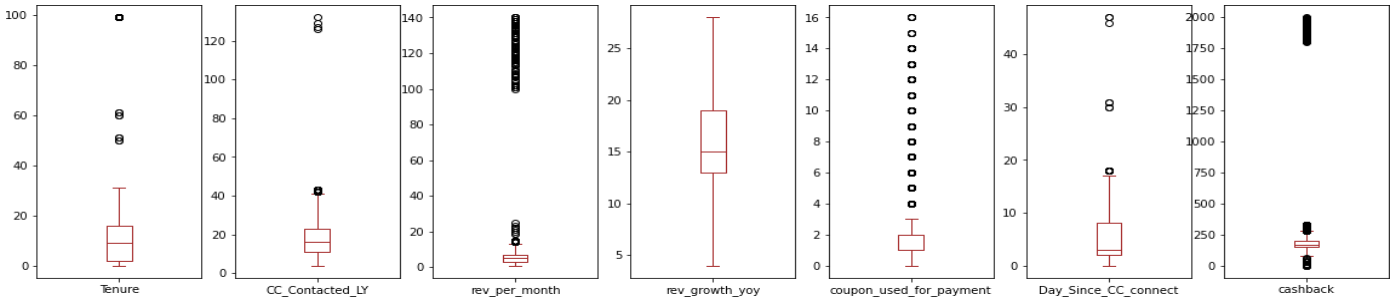
Univariate analysis (distribution and spread for every continuous attribute, distribution of data in categories for categorical ones):

Distribution and spread of continuous variables:

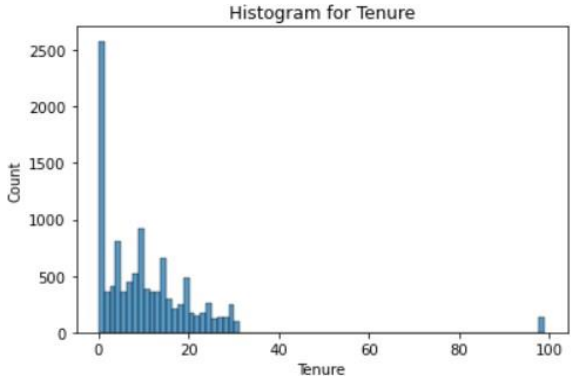
Univariate analysis is done for the purpose of observing distribution and spread for every continuous attribute and distribution of data in categories for categorical ones. It has been done by observing:

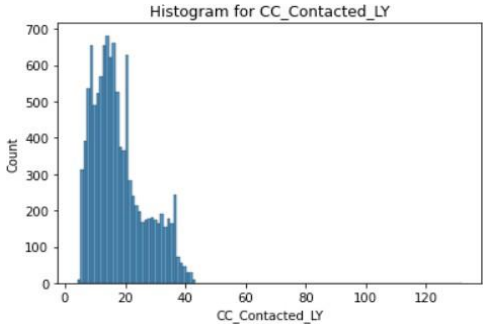
- Box plots and histograms for continuous variables
- Count plots for categorical variables

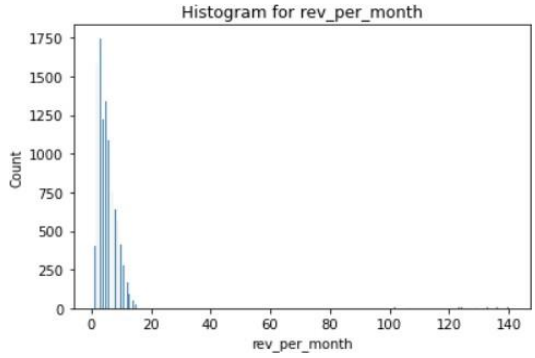
1) Box plot for continuous data :

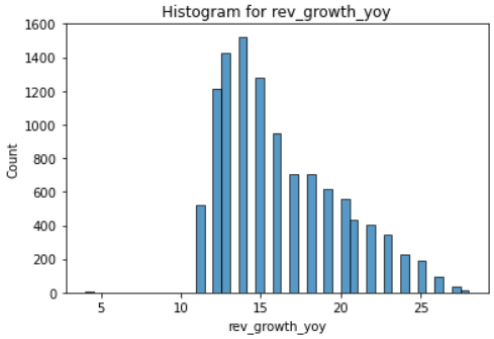


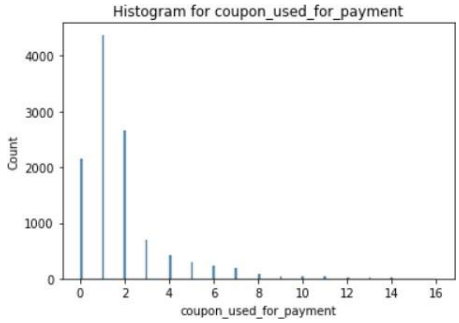
Continuous data – Histogram

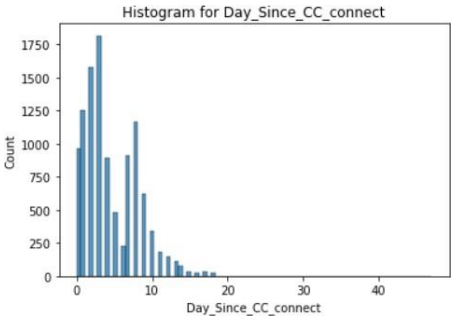












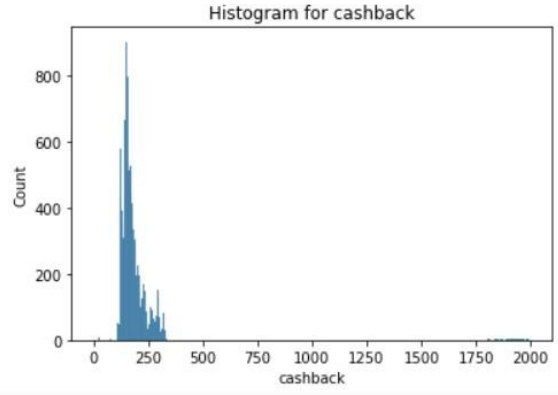


Table 3-1 Skewness of numeric variables

	Skewness
rev_growth_yoy	0.75
Day_Since_CC_connect	1.27
CC_Contacted_LY	1.42
coupon_used_for_payment	2.58
Tenure	3.90
cashback	8.77
rev_per_month	9.09

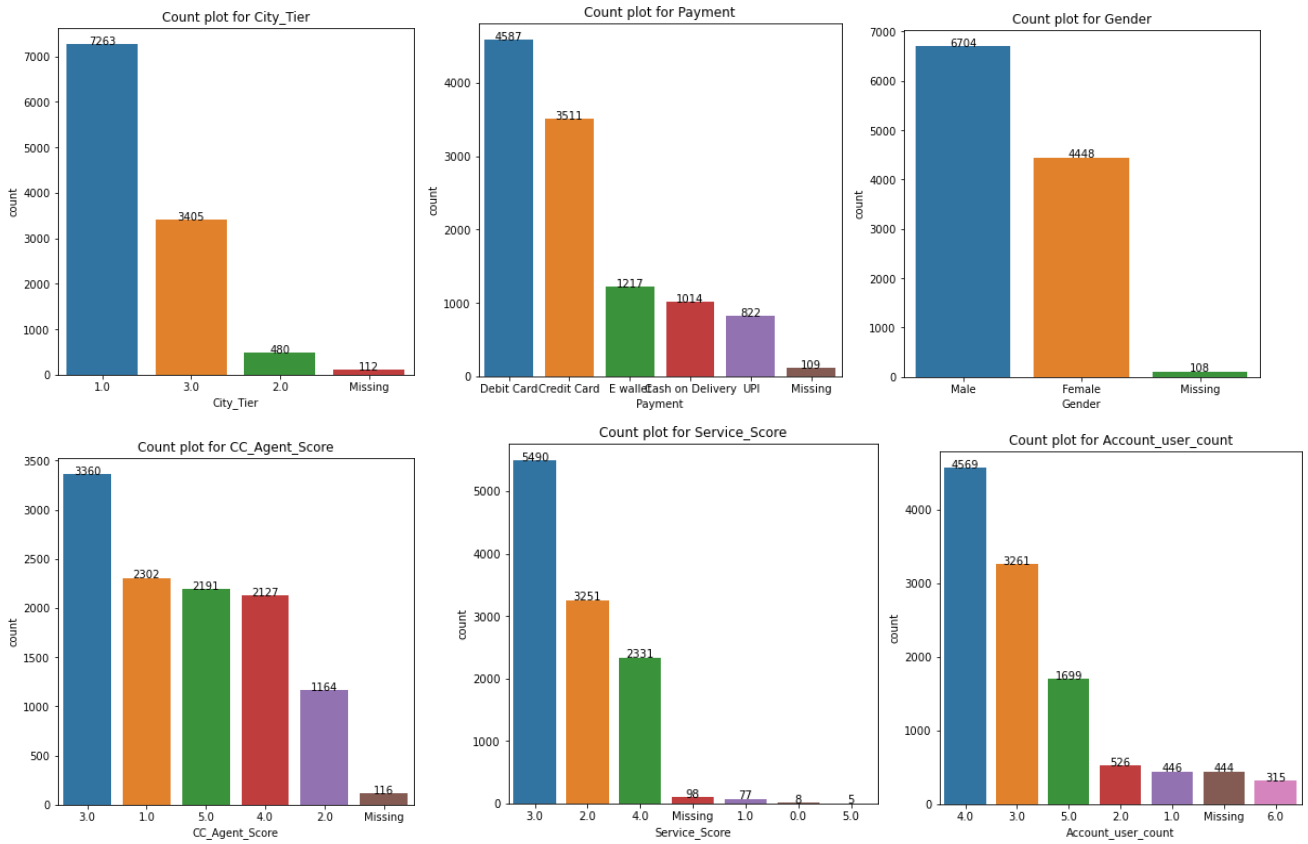
Observations:

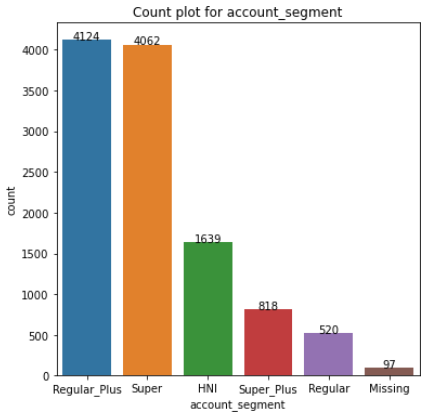
- All numeric variables with the exception of rev_growth_yoy have outliers. Some outliers for certain variables are closer to the whisker, whereas there are a group of outliers that are far beyond the whisker with no in between values. For instance, rev_per_month has a huge space between 30 and 100 indicating absence of values in that range. Those outliers in the extreme values do not correlate with corresponding outliers in cashback field. We cannot rule out these outliers as incorrect values, they may belong to hotels with many rooms. But models like logistic regression are sensitive to outliers and may not give good performance if outliers are left untreated.
- Hence, we will follow two approaches to modelling – one set of data with outliers treated for outlier sensitive models and other set of data with outliers not treated.
- Coupon_used_for_payment has a very limited range 0 to 16. Hence, for the purpose of this analysis, the outliers will not be treated (similar to categorical variables).
- All numeric variables with the exception of rev_growth_yoy have a high positive skew.

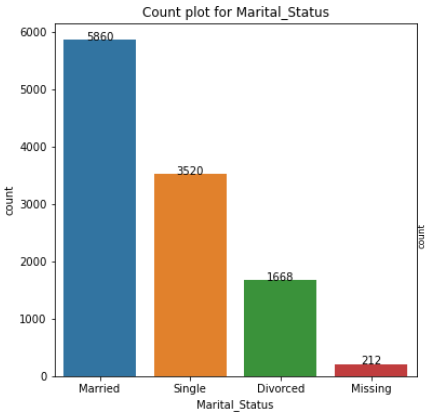
Below is the skewness for the above continuous variables:

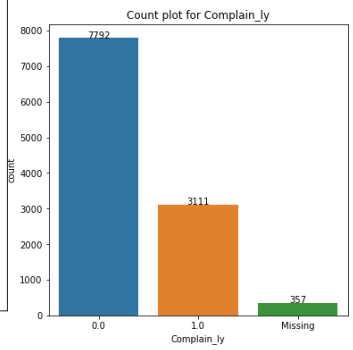
Now, let us see the categorical distribution plot:

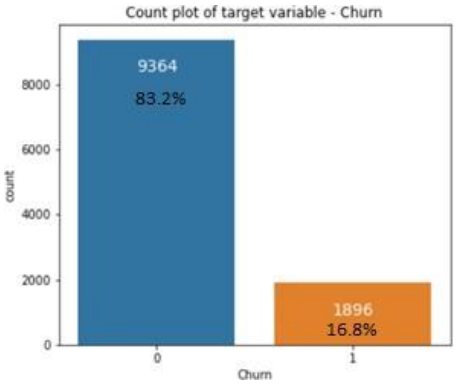
Categorical fields – Count plot



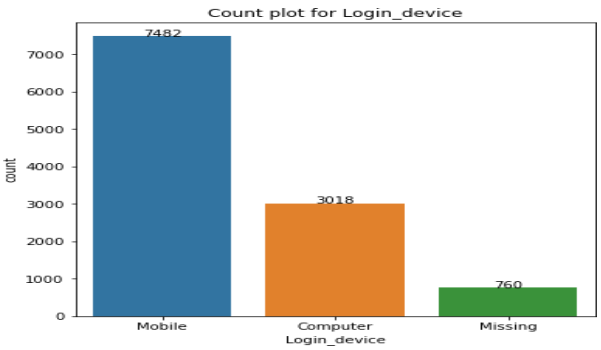








Observation: This is an imbalanced dataset with target variable containing 16.8% churn. The prediction will be for minority class.



Observations:

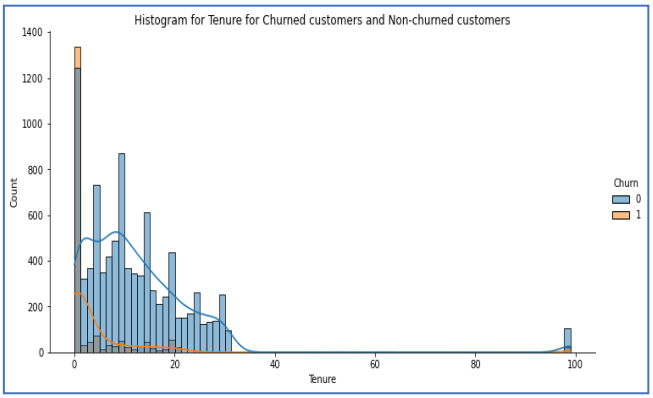
- This is an imbalanced dataset with target variable containing 16.8% churn
- Tier 1 cities have more accounts followed by Tier 3 cities
- Most of the accounts pay through debit card followed by credit card. UPI ranks last amongst payment methods
- Number of male account holders outnumber females
- Regular plus and Super are top two account segment types by number
- Top score for both Customer service agent and Service score is 3
- Married customers have the most accounts followed by single
- Most accounts do not have a customer complaint filed last year
- Most account holders use Mobile for logging and using services

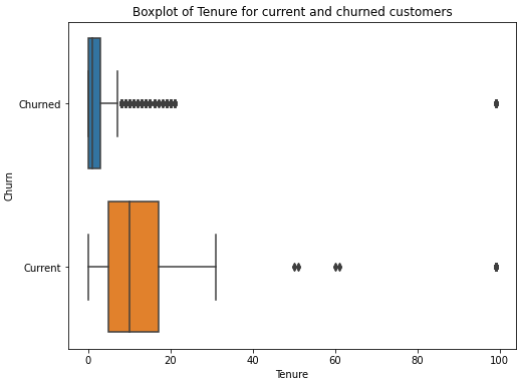
Bi -Variate Analysis:

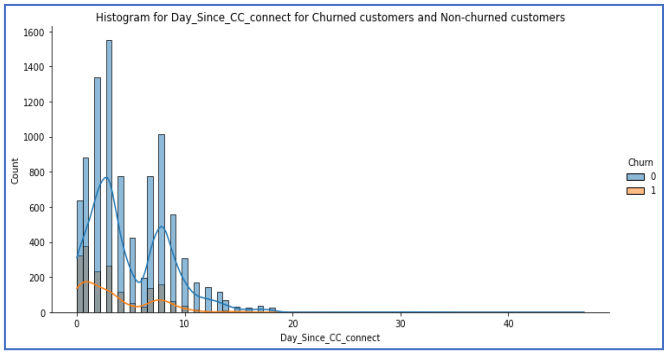
Bivariate analysis & Multivariate analysis

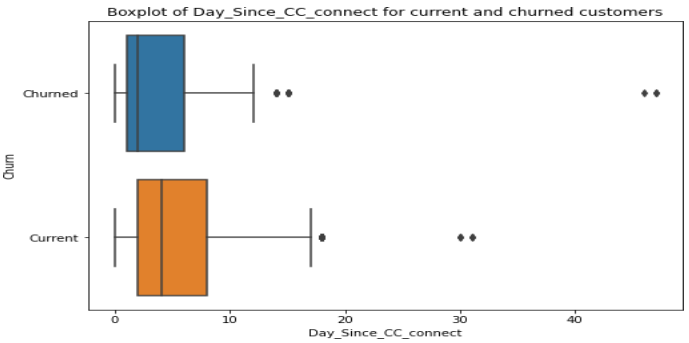
Bivariate analysis shows the relationship between two variables. Here, the predictor variables have been taken and their relationship with the target variable has been plotted. The influence of the predictor variables on the target variable can be observed in these bivariate plots.

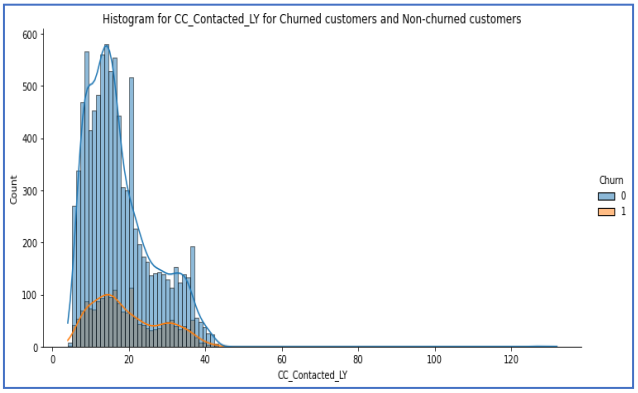
Continuous predictor variables that show some ability to separate the target variables

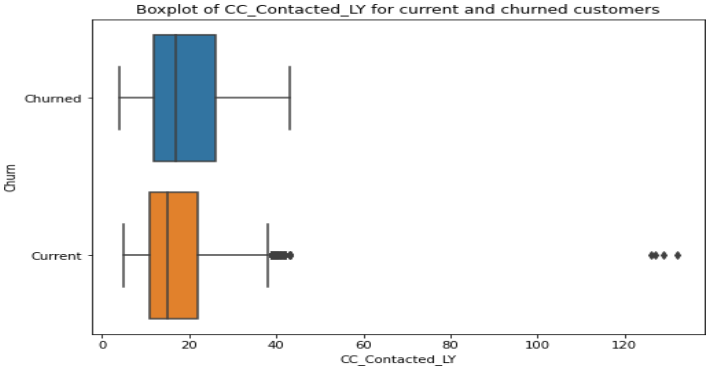




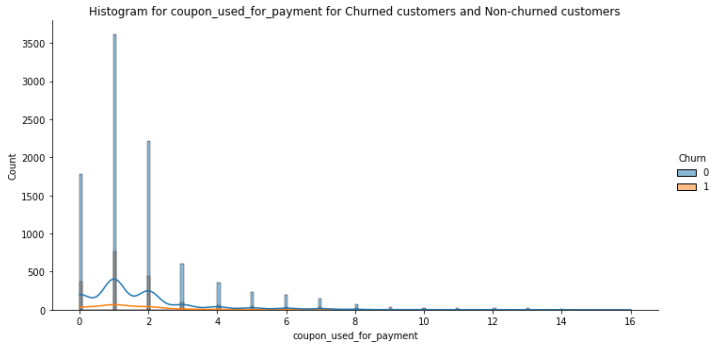


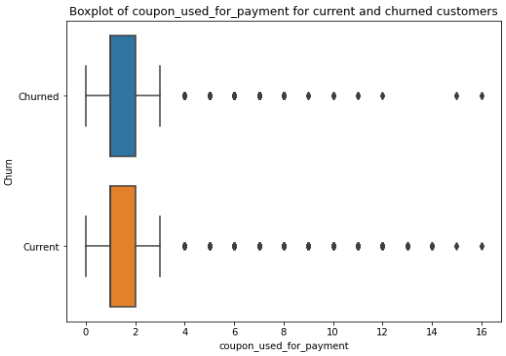


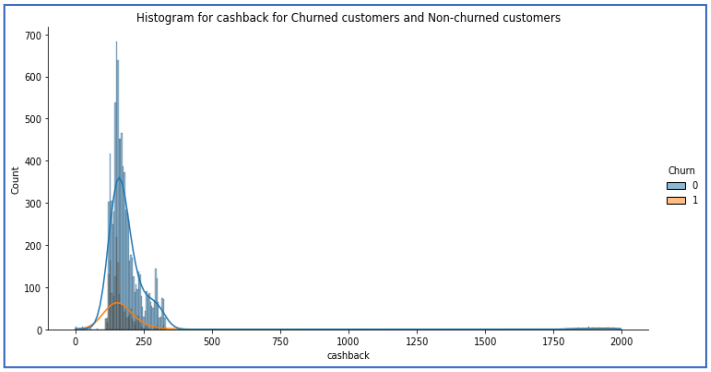


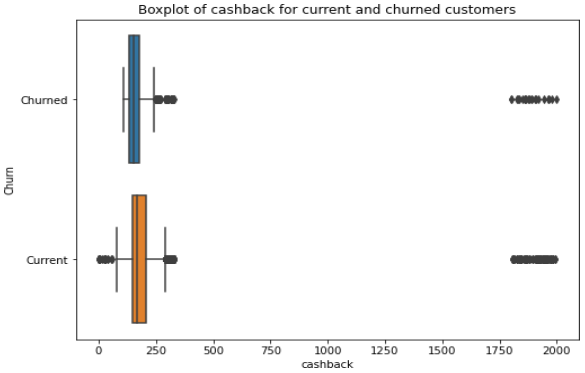


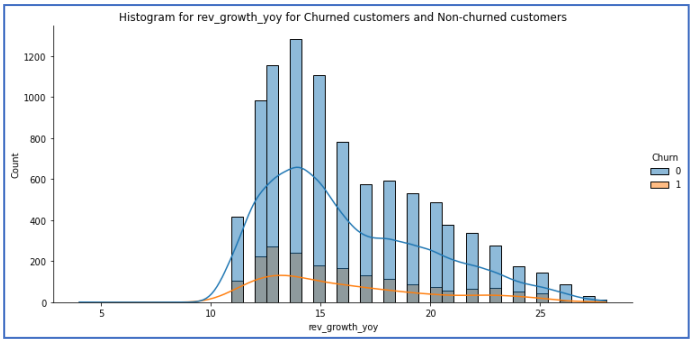
Continuous predictor variables that aren't able to show a clear separation between target classes

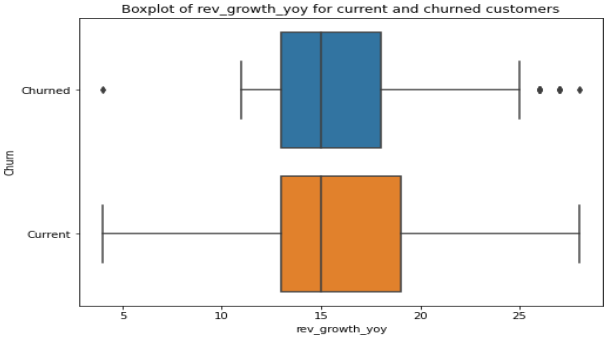












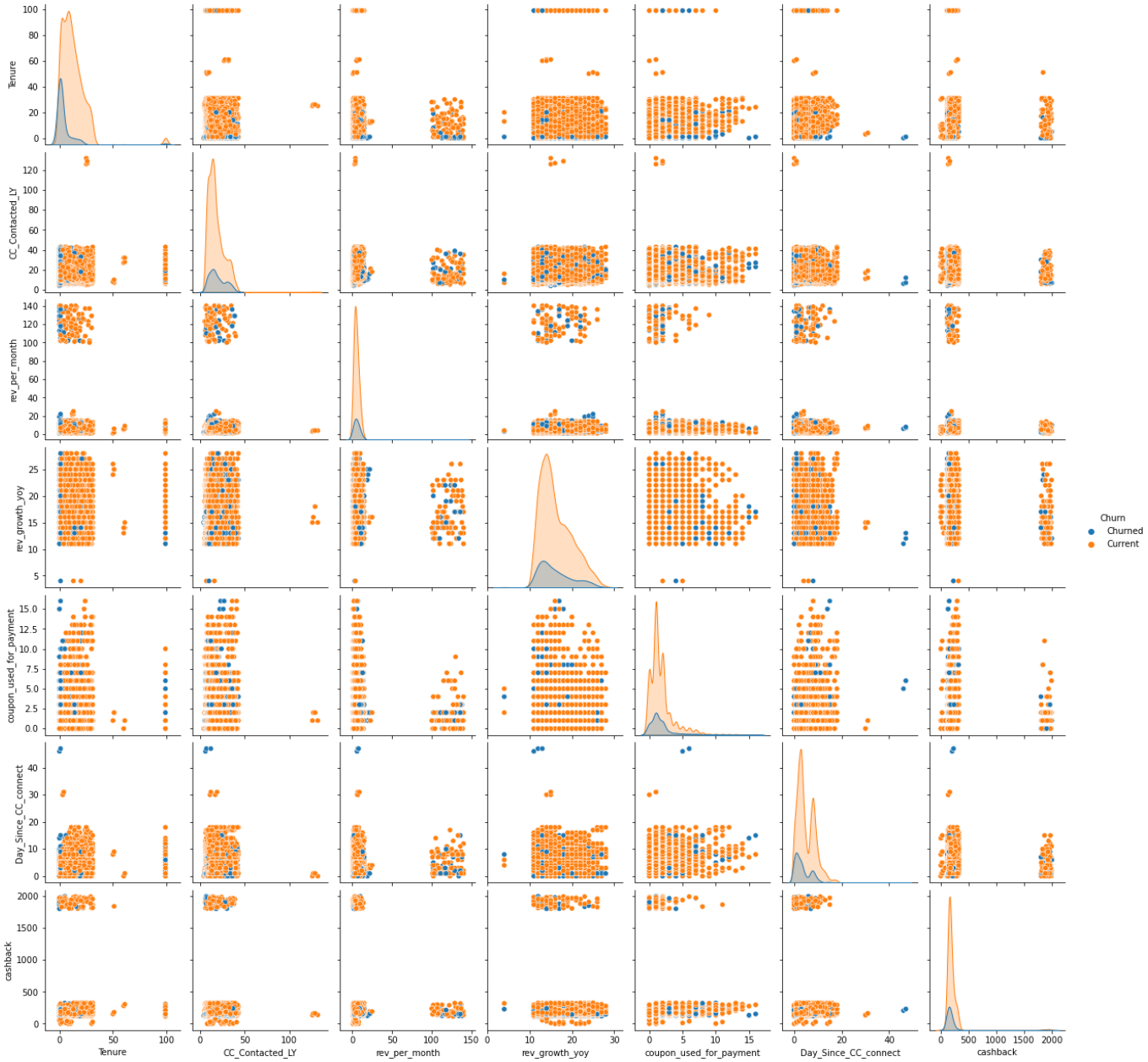
Observations: From the above plots we can see that variables such as Tenure, Days_since_CC_connect have some influence on the target variable. The median lines for Churned and Non-churned observations when plotted against these variables show a difference. Whereas, the medians in boxplots for variables such as coupon_used_for_payment, rev_growth_yoy do not show much difference in churned vs non-churned distributions.

Pair plot for the numeric variables with hue set as target variable

Observations:

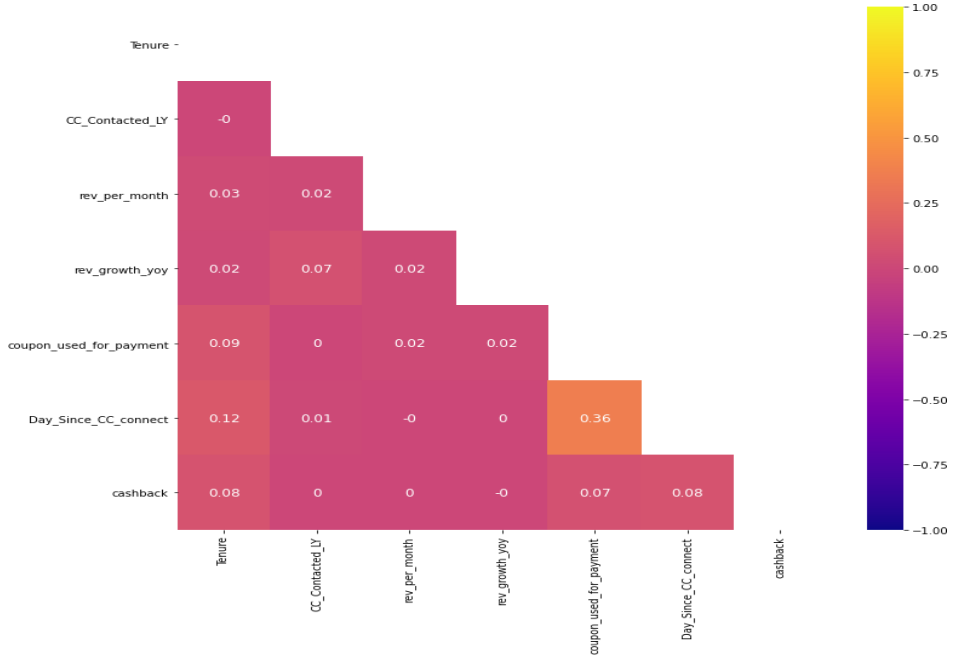
- Some of the variables clearly show presence of some clusters for e.g., rev_per_month and Day_Since_CC_connect
    - The diagonal kde plot for Tenure shows a slight separation with customers who churned falling on the lower side of Tenure

There is no linear relationship between any two continuous variable



Correlation heat map

The following heat map shows the correlation (pearson's) between various numeric predictors in the dataset. The purpose of doing a correlation matrix is to check if any of the variables have a strong correlation that can contribute to multi-collinearity. If there is a strong correlation > 0.80 between two variables, one of them can be dropped as it would not add much to the model's performance.



Observation: None of the numeric variables show a strong correlation. Hence there is no need to drop any variable

Removal of unwanted variables (if applicable)

The following are the checks done to see if any columns can be dropped before modelling exercise:

- Any variable that has unique values for each observation – for e.g., AccountID field. This would not contribute to the model as it is just an ID field to tag each observation.
- Any variable that remains constant for all or most of the observations as this does not add any strength to prediction. As observed from the histogram (numeric) and count plots/value counts (categorical), there are no variables that have constant value for all observations.
- Any variable that has nulls in more than 25 to 30% of observations. The maximum nulls present are in cashback variable and that column contains 4% nulls. Hence no column will be dropped.
- Any predictor variable that has a strong correlation with another predictor variable. Then one of the variables can be dropped. As seen in the correlation heat map above, there are no strong correlations, hence no variable needs to be dropped.
- Any predictor variable that has a very weak correlation with the target variable., two variables – rev_growth_yoy and coupon_used_for_payment are found to not be needed as they don't focus on providing much information related to the churning. Hence these two variables would be dropped from further processing.

Three variables (Cashback, Service score, Cluster code) were dropped as part of VIF check in later step

AccountID, rev_growth_yoy and coupon_used_for_payment have been removed

Addition of new variables

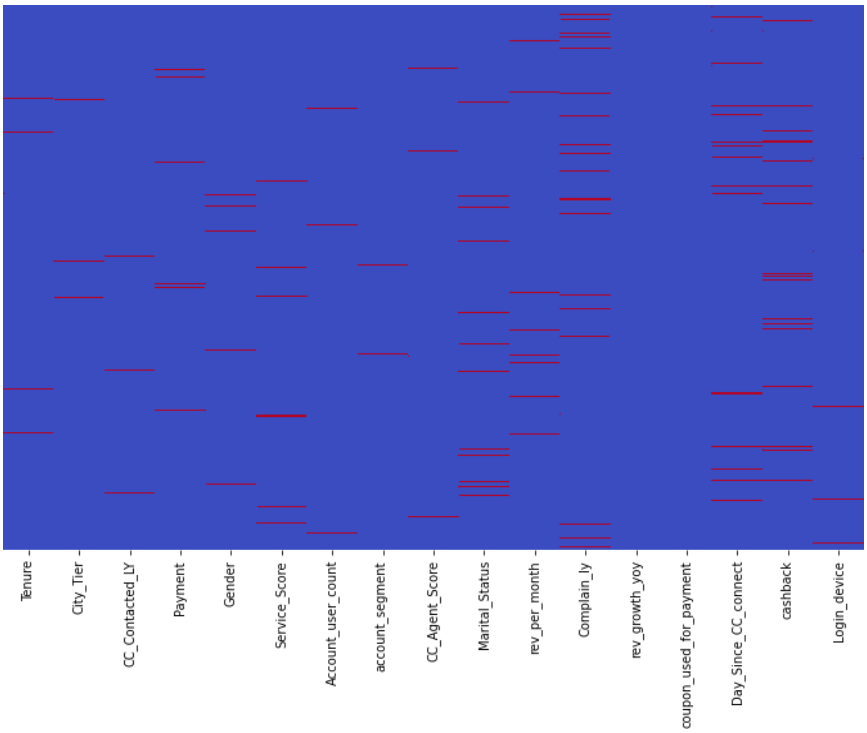
Cluster code has been added to the dataset (details about clustering is provided in section -Business report insights) . It may be used when experimenting with model building.

b)Missing valuetreatment

Percentage of nulls:

Percentage nulls or missing values present in the predictor variables of the dataset are as follows:

cashback	4.18
Complain_ly	3.17
Day_Since_CC_connect	3.17
Login_device	1.96
Marital_Status	1.88
CC_Agent_Score	1.03
Account_user_count	0.99
City_Tier	0.99
Payment	0.97
Gender	0.96
rev_per_month	0.91
CC_Contacted_LY	0.91
Tenure	0.91
Service_Score	0.87
account_segment	0.86
rev_growth_yoy	0.00
coupon_used_for_payment	0.00



Missing value treatment:

Missing value treatment was done using KNN imputation, a distance-based method. The following treatments were done as they are pre-requisites for missing value treatment using KNN.

- All variables need to be numeric. Any object-type categorical variables need to be encoded suitably (label/ one-hot encoding). The following columns were one hot encoded as it contained nominal categorical variables. The first encoded variable was dropped to avoid multi-collinearity

One hot encoded variables Payment','Gender','account_segment','Marital_Status','Login_device'

- After encoding, the column names were renamed in order to shorten, make it uniform and remove blank spaces that resulted out of some of the category values from one-hot encoded columns (e.g., Payment_Credit Card)
- All variables need to be scaled as KNN is a distance-based algorithm. Scaling was done using Standard Scaler function from SKLearn library for the predictor variables. The target variable was left as-is.
- Null imputing was done using Sklearn's KNNImputer function. This algorithm imputed missing values using K-nearest neighbors.

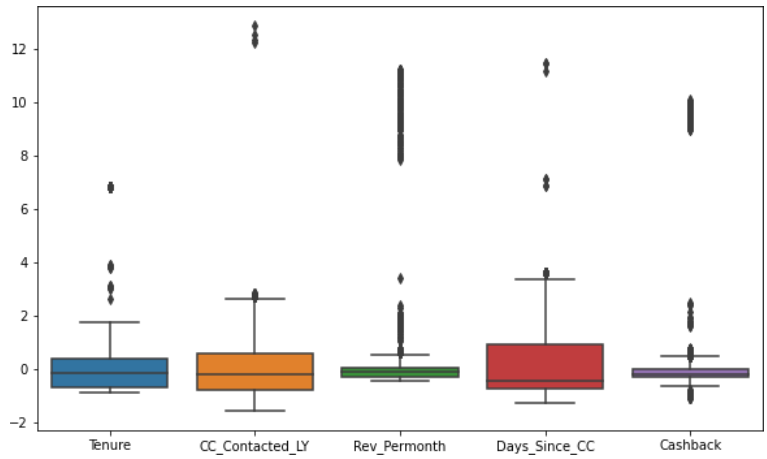
```
Tenure                      0
City_Tier                   0
CC_Contacted_LY             0
Service_Score               0
User_Count                  0
CC_Score                    0
Rev_Permonth                0
Complain_LY                 0
Days_Since_CC               0
Cashback                    0
Payment_Creditcard          0
Payment_Debitcard           0
Payment_Ewallet             0
Payment_UPI                 0
Gender_Male                 0
ACSegment_Regular           0
ACSegment_Regularplus       0
ACSegment_Super             0
ACSegment_Superplus         0
Maritalstatus_Married       0
Maritalstatus_Single        0
Logindevice_Mobile          0
```

Variable transformations:

- **Encoding:** As mentioned in the above section, the variables **Payment, Gender, Account_Segment, Marital_Status and Login_device** are all categorical object type.
- They need to be converted to numeric variables. The categories in these variables do not have an order. Hence, they were one-hot encoded. When performing one-hot encoding, the variable is split into multiple columns with each column taking binary values corresponding to each category in the original attribute. Additionally, one of the columns in the one-hot encoded variables is dropped in order to avoid multi-collinearity which may cause performance degradation and interpretability issues in some of the models.
- **Scaling:** The dataset had been scaled as it is a pre-requisite for any distance-based algorithm like K-means clustering. The scaled data can also be used by all models irrespective of whether they expect scaled input or not.
- No other transformation is expected for modeling as of now.

c) Outlier treatment

The following continuous variables have outliers.



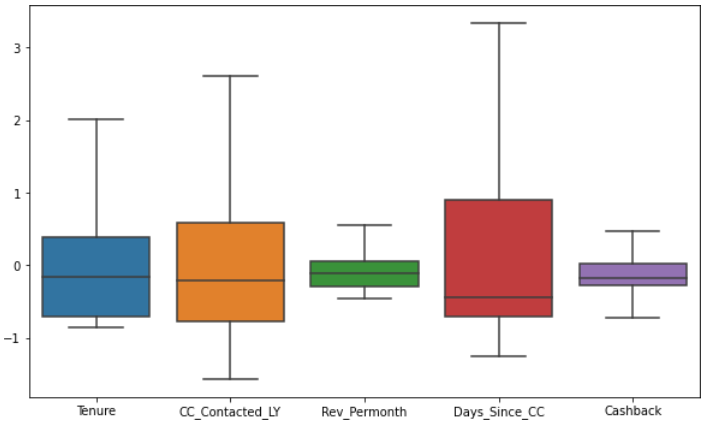
Some outliers for certain variables are closer to the whisker, whereas there are a group of outliers that are far beyond the whisker with no in between values. For instance, rev_per_month has a huge space between 30 and 100 indicating absence of values in that range. Those outliers in the extreme values do not correlate with corresponding outliers in cashback field. We cannot rule out these outliers as incorrect values, they may belong to hotels with many rooms. But models like logistic regression are sensitive to outliers and may not give good performance if outliers are left untreated.

- Hence, two approaches to modelling were performed – one set of data with outliers treated for outlier sensitive models and other set of data with outliers.
- Coupon_used_for_payment has a very limited range 0 to 16. Hence, for the purpose of this analysis, the outliers will not be treated (similar to categorical variables).

For the outlier treated dataset, outliers beyond upper and lower whiskers were treated by capping to the lower and upper range where

- lower_range= 1st quartile - (1.5 * IQR) and
- upper_range= 3rd quartile + (1.5 * IQR)

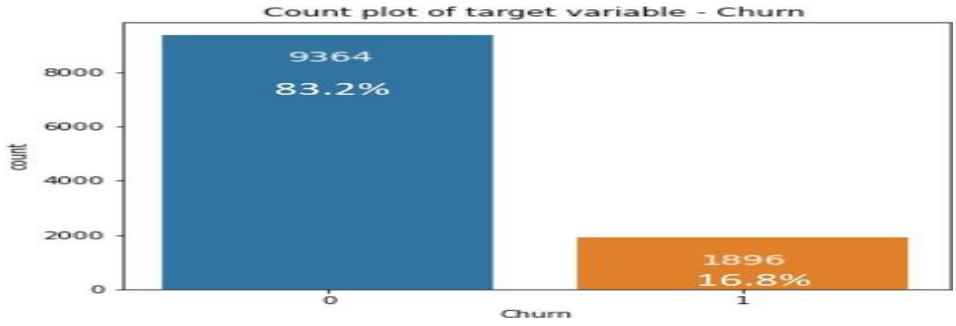
Where, IQR = 3rd quartile – 1st quartile value



	Skewness
Tenure	0.80
CC_Contacted_LY	0.80
Rev_Permonth	0.78
Complain_LY	0.95
Days_Since_CC	0.82
Cashback	0.93

Data imbalance

As can be seen from the plot below, the data is imbalanced with respect to the target variable which is the indicator whether customer has churned or not. The distribution given below shows that for every 100 customers acquired by the business, 17 have churned and 83 customers are active. This distribution is skewed towards active/current customers. The objective for this exercise is to be able to predict the customers who would churn i.e., the minority class '1'



If a dataset has equal distribution amongst the categorical values taken by the target variable (Churn), it would be called a balanced dataset. In a balanced dataset, the model learns to predict both classes with equal efficiency as there are equal number of observations.

In case of customer churn, any business would have less of churned customers and more of active customers. Similarly, this dataset also has close to 17% churned customers. Although this mimics real-life churn data, from a modelling perspective, this may pose a challenge.

Challenges due to imbalance dataset:

- If there are no sufficient observations in the minority class, the model would be unable to learn the patterns in minority class (class of interest) well and may predict majority class better. Using resampling techniques such as oversampling minority class or under-sampling majority class may be beneficial. Compared to under-sampling majority class which would result in loss of data, oversampling using techniques like SMOTE (Synthetic Minority Oversampling) may help. This technique would help generate synthetic data for churned customers based on the existing churned customer observations. However, this technique may not always guarantee better model performance. Depending on the algorithm used and the type of SMOTE used, there could be overfitting in the train dataset.
- Accuracy as an evaluation metric would not be appropriate in imbalanced classification problems. Even if the algorithm predicts all customers as belonging to majority class, it would still result in an accuracy of 83.2%. Using Precision, Recall or F1-score for the minority class may be a better approach for evaluation.

  In this problem, SMOTE would be one of the data treatments given. Models built on imbalanced data would be compared against models built on SMOTE balanced dataset and the best performance measure yielding data would be chosen. It may be possible that SMOTE does not give a good performance, hence a comparison would be required to make that decision.

Clustering

- **Cluster analysis** or **clustering** is the task of grouping a set of objects in such a way that objects in the same group (called a **cluster**) are more similar to each other than to those in other groups (clusters).
- Clustering is an unsupervised task and given all the features in the dataset, the clustering algorithm is allowed to group customers such that each group has similar customers and customers of different groups are dissimilar.For this purpose, the processed dataset (cleaned, scaled, nulls imputed, outlier treated, categorical features encoded) without the target variable was used.
- As the dataset contained both continuous and categorical predictor variables, kprototypes function from Kprototypes library was used.The algorithm was run for 2,3 and 4 clusters and 3 clusters seemed to have

the best separation. The cluster profile was formed by grouping the observations by clusters and finding the mean for all the features.

- Although churn was not part of the features for clustering, it was added part of the cluster profile so that it is possible to appreciate how churn varies for each cluster.

Clusters	Churn	Tenure	City_Tier	CC_Contacted_LY	Complain_LY	Days_Since_CC	Cashback	ACSegment_Regularplus	ACSegment_Super
1	0.224359	10.63406	1.679252	30.46978335	0.312962242	3.826426896	194.014391	0.328808446	0.425339367
0	0.187367	9.499803	1.599688	13.29957065	0.292197452	2.218893549	183.1879527	0.540853776	0.306355032
2	0.096473	13.62615	1.716176	15.00088158	0.25347432	8.874138448	217.4964325	0.131740017	0.392888371

The insights from clustering have been provided below:

Business Insights from cluster profiling

- Even though this is an unsupervised algorithm without the use of target variable in clustering, the profiling came up with clear separation of groups only for those features that also showed a high F-statistic and high Chi square value in the bivariate analysis.
- Higher the tenure, lesser the churn. But the cluster profile also shows that for low tenures (cluster 1 and 0), this is not holding good. To further explore this, it would be good to bin Tenure and check the relationship with Churn through a stacked bar.
- More Regular plus customers have a greater churn compared to least churn clusters
- High churn clusters had contacted customer care almost twice as the least churn customers
- The cashback was lower for high churn cluster compared to low churn cluster
- High churn cluster contacted customer care less times compared to low churn customers
- The cluster with maximum complaints last year also has the maximum churn. The cluster with minimum complaints last year has the minimum churn. It could be suggested that if any customer files a complaint, the complaint be followed up and resolved until the customer is satisfied. This process needs to be looked at if it can be strengthened.

MODEL BUILDING

Algorithms applicable for the given problem

- In this business case, the need is to predict whether a given customer would churn or not. This is a binary classification problem with only two prediction outcomes '0' – will not churn and '1' – will churn
- Since there is a target variable 'Churn' to be predicted, this is a supervised learning problem
- There are several algorithms that can be used for classification problems such as these
    - Linear classification: Logistic Regression, Linear Discriminant Analysis, Naïve Bayes
    - Non-linear classification algorithms: SVMs non-linear adaptations, Decision tree, K-nearest neighbor, Artificial Neural Network
    - Ensemble models: Rndom Forest, Adaboost, Gradient Boost
- These algorithms have certain assumptions about the data on which they are fit. Depending on the nature of the data, algorithms would give good or poor performance.

Approach

- Different treatments of pre-processed data were prepared - with and without outliers, with and without SMOTE resampling, with and without scaling.
- VIF (Variance Inflation Factor) was calculated for all the predictor variables. One by one predictor variables whose VIF was more than 5 were identified. 4 variables – Cashback, Service Score, Clusters and User count had VIF greater than 5. It is to be noted that even though User count had a high VIF value, it was retained as it showed significant Chi square value as part of EDA. The remaining 3 variables were dropped before processing the datasets; hence, none of the models have used these variables as predictors.

➢ Model Building and Tuning

3 algorithms were tried for this dataset – Logistic Regression, Linear Discriminant Analysis, Gradient Boost. Algorithm implementations from Sklearn package was used for all algorithms. In addition to SKlearn, Logistic regression algorithm was also executed using Statsmodel package. The Appendix contains detailed information for all the 3 models including details on the base and the tuned model performance along with their interpretations.

Effort for model tuning

Model performance improvement was accomplished using the following methods:

- Around 3 different algorithms were tried across linear, non-linear and ensemble methods.
- The first model for each algorithm was the base model with the default hyperparameters. Model tuning to achieve better performance was done by tuning the hyperparameters using Grid Search CV, a function offered by Sklearn to automatically try out multiple parameter options for each hyperparameter.
- The underlying data was changed (Smote resampled/ non-resampled, outlier treated/not treated) and the effect of model performance observed for some of the models.
- Ensemble methods were used (Gradient Boost) and their hyperparameters were also tuned.

The following tables show all the algorithms tried, various model tuning trials done and the performance for each one of them in train and test dataset. The data treatment done is also shown as part of the table.
Hyperparameters are part of the code and hence not included here. The best model for each algorithm has been highlighted in green. The model reference number can be used to locate the model in the python code.

1) Logistic Regression:

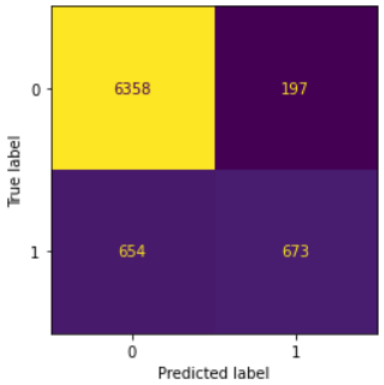
Logistic regression model performs well when the data is linearly separable. It assumes that there is linearity between target and predictor variables. It is a parametric model. It also provides coefficients that helps with model interpretability.

Hence the first model tried was Logistic regression. Both SKLearn and Statsmodel (pytho libraries) implementations were tried for Logistic regression.
Logistic regression is outlier sensitive; hence only outlier treated data has been used for all the models.

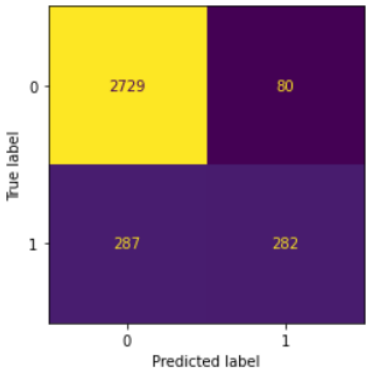
- SKlearn's Logistic Regression was used alongside RFE (Recursive feature elimination) to determine number of features that can give best performance and the ranking of features that Logistic regression provides.
- Statsmodel implementation was tried to obtain p-values to determine what features need to be retained in the model and to use the coefficients to understand the relationship of predictor and target variables

SK Learn Base model with default hyperparameters (Also the best model)

Train dataset Confusion Matrix



Test dataset Confusion Matrix

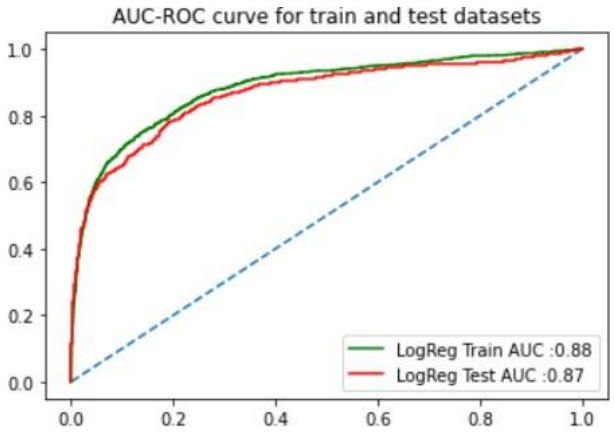


Train dataset Classification report

	precision	recall	f1-score	support
0	0.91	0.97	0.94	6555
1	0.77	0.51	0.61	1327
accuracy			0.89	7882
macro avg	0.84	0.74	0.77	7882
weighted avg	0.88	0.89	0.88	7882

Test dataset Classification report

	precision	recall	f1-score	support
0	0.90	0.97	0.94	2809
1	0.78	0.50	0.61	569
accuracy			0.89	3378
macro avg	0.84	0.73	0.77	3378
weighted avg	0.88	0.89	0.88	3378



It is to be noted that this is also the best model out of all Logistic Regression models

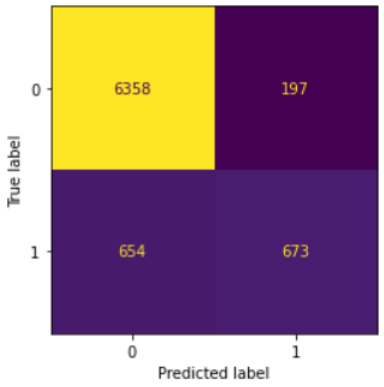
Statsmodel Logistic regression - model tuning

- Statsmodel provides a model summary when a model is fit on the train data. The p-value of the predictor variables in the summary was used for deciding the significance of each predictor variable to the model.
- One by one the predictor variables whose p-value was > 0.05 were removed from the model and model rebuilt. Over multiple iterations (as shown below), the variables were eliminated one by one such that only variables with p-value < 0.05 remained in the model. 16 variables were significant at a level of 0.05.

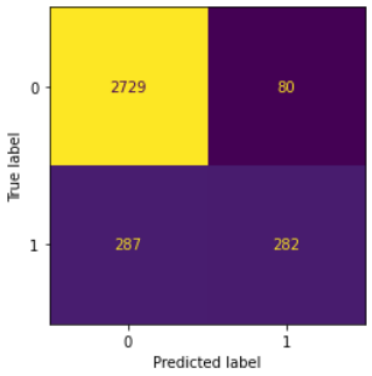
The F1-score of the final model built was close but not greater than the SkLearn's best model. Hence this was not selected as the best model in logistic regression.

SK Learn Base model with default hyperparameters (Also the best model)

Train dataset Confusion Matrix



Test dataset Confusion Matrix

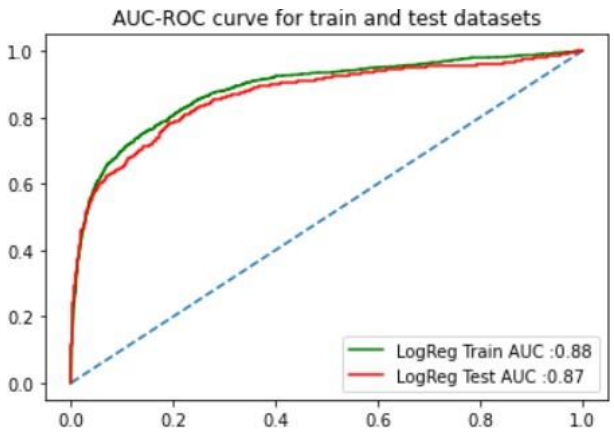


Train dataset Classification report

	precision	recall	f1-score	support
0	0.91	0.97	0.94	6555
1	0.77	0.51	0.61	1327
accuracy			0.89	7882
macro avg	0.84	0.74	0.77	7882
weighted avg	0.88	0.89	0.88	7882

Test dataset Classification report

	precision	recall	f1-score	support
0	0.90	0.97	0.94	2809
1	0.78	0.50	0.61	569
accuracy			0.89	3378
macro avg	0.84	0.73	0.77	3378
weighted avg	0.88	0.89	0.88	3378



It is to be noted that this is also the best model out of all Logistic Regression models

Statsmodel Logistic regression - model tuning

- Statsmodel provides a model summary when a model is fit on the train data. The p-value of the predictor variables in the summary was used for deciding the significance of each predictor variable to the model.
- One by one the predictor variables whose p-value was > 0.05 were removed from the model and model rebuilt. Over multiple iterations (as shown below), the variables were eliminated one by one such that only variables with p-value < 0.05 remained in the model. 16 variables were significant at a level of 0.05.

The F1-score of the final model built was close but not greater than the SkLearn's best model. Hence this was not selected as the best model in logistic regression.

SKLearn Logistic regression - model tuning

- The base model (Section 7.1.1) was further tuned by using RFE to change the number of predictors used. It was found that reducing variables did not yield better results. Hence all 20 predictors were retained.
- GridSearchCV function from Sklearn library was used to tune the hyperparameters. It was found that tuned hyperparameters did not perform better than the base model.
- Further, Scaled data and Smote resampled data was used with the base model. It was found that Smote resulted in overfitting precision of class 1. Scaling also did not improve performance compared to non-scaled data.
- The results of all the trials have been given in the table below
- The best model is the same as the base model mentioned which is LR_model1 (reference to the python code) highlighted in green in the table below

Model reference	Data treatment		Hyper parameters	Train data					Test data				
	Smote	Scaling		Accuracy	Precision	Recall	F1	AUC	Accuracy	Precision	Recall	F1	AUC
LR_model1	**No**	**No**	**Default base model**	**0.89**	**0.77**	**0.51**	**0.61**	**0.88**	**0.89**	**0.78**	**0.5**	**0.61**	**0.87**
LR_model2	No	No	Default base model, RFE variables=8	AUC = 0.77					AUC = 0.77				
LR_model3	No	No	Default base model, RFE variables =12	AUC = 0.78					AUC = 0.78				
LR_model4	No	No	Default base model, RFE vars=16	0.89	0.78	0.49	0.6	0.88	0.89	0.8	0.47	0.59	0.87
LR_model5	No	No	Default base model, RFE vars=18	0.89	0.77	0.49	0.6	0.88	0.89	0.79	0.48	0.6	0.87
LR_model6	No	No	Default base model, RFE vars=19	0.89	0.77	0.49	0.6	0.88	0.89	0.8	0.48	0.6	0.87
LR_model7	No	No	Gridsearch CV, best model for f1 score	0.89	0.77	0.5	0.61	0.88	0.89	0.78	0.49	0.6	0.87
LR_model8	Yes	No	Default base model	0.81	0.8	0.82	0.81	0.89	0.79	0.44	0.79	0.56	0.87
LR_model9	No	Yes	Default base model	0.89	0.77	0.5	0.61	0.88	0.89	0.78	0.49	0.6	0.87
LR_model10 **Statsmodel**	No	No	Iterated 4 times to remove 4 variables	0.89	0.78	0.51	0.61	0.88	0.89	0.79	0.48	0.6	0.87

Model interpretation

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	-4.3795	0.264	-16.616	0.000	-4.896	-3.863
Tenure	-0.1911	0.008	-24.653	0.000	-0.206	-0.176
City_Tier	0.3733	0.045	8.276	0.000	0.285	0.462
CC_Contacted_LY	0.0330	0.004	7.342	0.000	0.024	0.042
User_Count	0.3469	0.039	8.801	0.000	0.270	0.424
CC_Score	0.2592	0.028	9.114	0.000	0.203	0.315
Rev_Permonth	0.1484	0.013	11.438	0.000	0.123	0.174
Complain_LY	1.6698	0.079	21.029	0.000	1.514	1.825
Days_Since_CC	-0.0686	0.012	-5.534	0.000	-0.093	-0.044
Payment_Creditcard	-0.6387	0.111	-5.756	0.000	-0.856	-0.421
Payment_Debitcard	-0.4482	0.102	-4.382	0.000	-0.649	-0.248
Payment_UPI	-0.5182	0.158	-3.280	0.001	-0.828	-0.209
Gender_Male	0.2829	0.078	3.609	0.000	0.129	0.437
ACSegment_Regular	1.0364	0.268	3.867	0.000	0.511	1.562
ACSegment_Super	-1.2345	0.094	-13.197	0.000	-1.418	-1.051
Maritalstatus_Single	0.9244	0.078	11.860	0.000	0.772	1.077
Logindevice_Mobile	-0.3870	0.079	-4.887	0.000	-0.542	-0.232

This model's performance with respect to precision and recall of minority class is not a good one. A recall of 0.50 means out of 100 churning customers, model can only identify 50 correctly. The precision is not very high either. So other models may have to be looked at for this problem. Since the model has not performed well, the coefficients are only an academic exercise and business cannot rely on that for insights.

The following variables have a positive correlation with Churn: i.e., as the variable increases, churn increases:
City tier, User count, Customer care score, Revenue per month, Complaint last year, Regular customer segment and Single marital status.

The following variables have a negative correlation with Churn, i.e, as the variable increases, churn decreases:

Tenure, Credit/debit/UPI payment, Super segment, Mobile login device

Unlike linear regression models, the coefficients become non-intuitive to interpret in logistic regression models. For instance, the logistic regression coefficient for City_Tier means: as City_tier goes up by 1, the log odds of churning go up by 0.37. It has been broken down for the top 8 influential variables below:

From the table above, the logistic regression equation for this model is as follows:

logit(p) or log(p/1-p) or log(P{Y=1}/P{Y=0}) or log-odds ratio =

*-4.38 – 0.19*Tenure + 0.37*City_tier + 0.03*CC_contacted_LY + 0.35*User_Count + 0.26*CC_Score + 0.15*Rev_Permonth + 1.67*Complain_LY – 0.07*Days_Since_CC – 0.64*Payment_Creditcard – 0.45*Payment_Debitcard – 0.52*Payment_UPI + 0.28*Gender_Male + 1.04*ACSegment_Regular – 1.23*ACSegment_Super + 0.92*Maritalstatus_Single – 0.39*Logindevice_Mobile*

Coefficient	Predictor	Exponential of coefficient/ Odds of customer churning	Interpretation
1.67	Complain_LY	5.31	If complaint last year increases by 1, the odds of customer churning increases by 531%
-1.23	ACSegment_Super	3.42	If Account segment is Super (increases by 1), the odds of customer churning decreases by 342%
1.04	ACSegment_Regular	2.83	If Account segment is Regular (increases by 1),

			the odds of customer churning increases by 283%
0.92	Maritalstatus_Single	2.51	If Marital status is single (increases by 1), the odds of customer churning increases by 251%
-0.64	Payment_Creditcard	1.90	If payment is done by credit card (increases by 1), the odds of customer churning decreases by 190%
-0.52	Payment_UPI	1.68	If payment is done by UPI (increases by 1), the odds of customer churning decreases by 168%
-0.45	Payment_Debitcard	1.57	If payment is done by debit card (increases by 1), the odds of customer churning decreases by 157%
-0.39	Logindevice_Mobile	1.48	If login device is mobile (increases by 1), the odds of customer churning decreases by 148%

As we can see from above, Tenure which from EDA seemed one of the significant separators of churn and non-churn customers does not figure in the top 8 influential variables. The model performance in terms of precision, recall and F1 score are not good compared to other models (non-linear and ensemble).

LR_model1 is the best model out of the Logistic regression models.

. Linear Discriminant Analysis:

Linear Discriminant Analysis (LDA) is also another machine learning classifier. It works well when there are linearly separable classes in data. It assumes that the underlying data has a gaussian distribution but can perform well even if assumptions are violated.

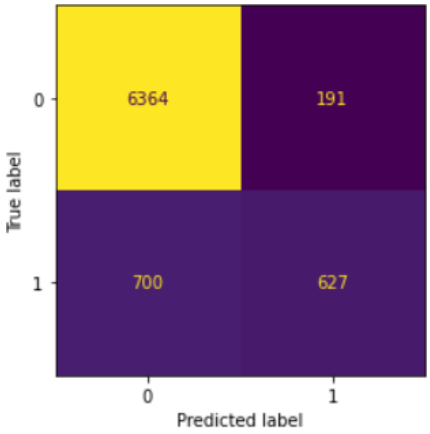
The detailed information regarding base model, best performing model, their train-test metrics, model interpretability are provided in below

Linear Discriminant Analysis (LDA) is also another machine learning classifier. It works well when there are linearly separable classes in data. It assumes that the underlying data has a gaussian distribution but can perform well even if assumptions are violated.

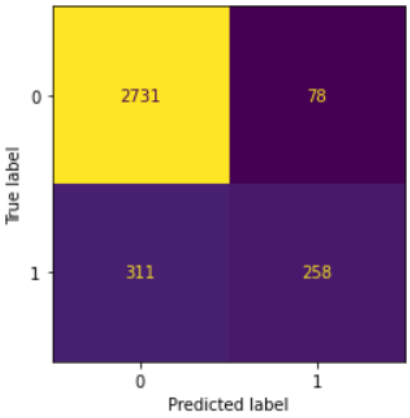
- SKlearn's Linear Discriminant Analysis function was used for modelling
- The base model was run with default hyperparameters and the performance metrics noted
- The model's hyperparameters were tuned using GridSearchCV function and model constructed using the best parameters selected by GridSearchCV. This did not provide much improvement over the base model except that f1-score improved by 0.01
- Data used was outlier treated unscaled dataset
- Model performance metrics for base model and tuned model have been provided below

Linear Discriminant Analysis model with default hyperparameters

Train dataset Confusion Matrix



Test dataset Confusion Matrix

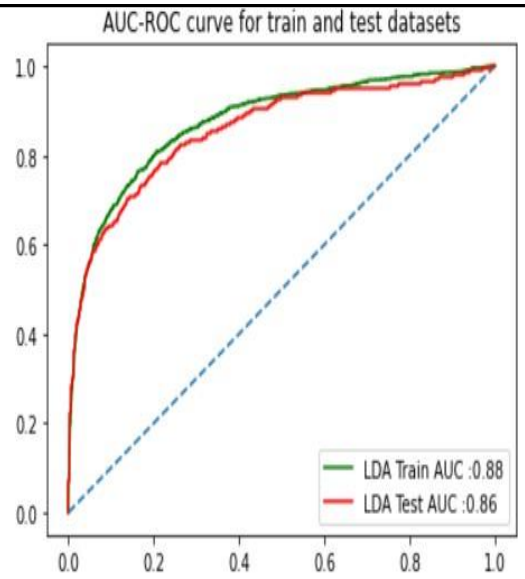


Train dataset Classification report

	precision	recall	f1-score	support
0	0.90	0.97	0.93	6555
1	0.77	0.47	0.58	1327
accuracy			0.89	7882
macro avg	0.83	0.72	0.76	7882
weighted avg	0.88	0.89	0.88	7882

Test dataset Classification report

	precision	recall	f1-score	support
0	0.90	0.97	0.93	2809
1	0.77	0.45	0.57	569
accuracy			0.88	3378
macro avg	0.83	0.71	0.75	3378
weighted avg	0.88	0.88	0.87	3378



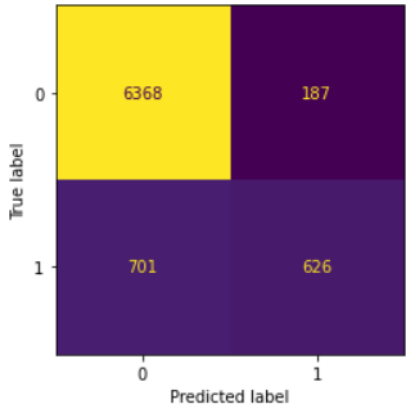
SKLearn LDA - model tuning

- GridSearchCV function from Sklearn library was used to tune the hyperparameters on the base model described was found that tuned hyperparameters performed almost similar to the base model.
- The results of all the trials have been given in the table below
- The best model is highlighted in green in the table below and the metrics for this model is (hyperparameter tuned model)

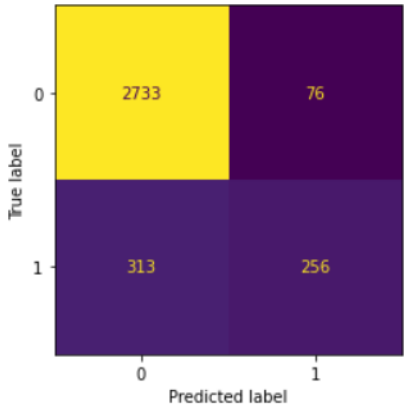
Data Used			Hyper	Train data					Test data				
Outliers	Smote	Scaling	parameters	Accuracy	Precision	Recall	F1	AUC	Accuracy	Precision	Recall	F1	AUC
Treated	No	No	Default base model	0.89	0.77	0.47	0.58	0.88	0.88	0.77	0.45	0.57	0.86
Treated	**No**	**No**	**Gridsearch CV, best model for f1 score**	**0.89**	**0.77**	**0.47**	**0.59**	**0.88**	**0.88**	**0.77**	**0.45**	**0.57**	**0.86**

LinearDiscriminantAnalysismodelwithGridsearchtunedhyperparameters

Train dataset Confusion Matrix



Test dataset Confusion Matrix

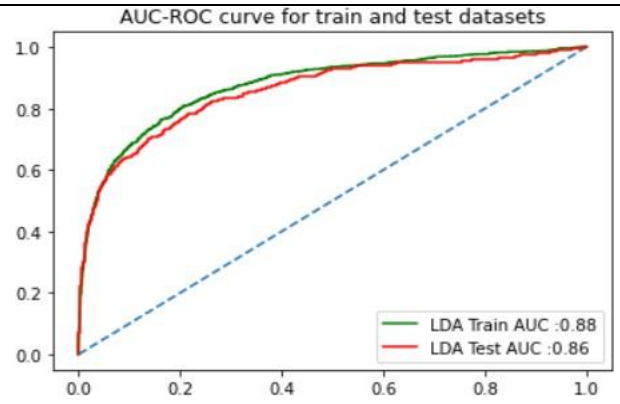


Train dataset Classification report

	precision	recall	f1-score	support
0	0.90	0.97	0.93	6555
1	0.77	0.47	0.59	1327
accuracy			0.89	7882
macro avg	0.84	0.72	0.76	7882
weighted avg	0.88	0.89	0.88	7882

Test dataset Classification report

	precision	recall	f1-score	support
0	0.90	0.97	0.93	2809
1	0.77	0.45	0.57	569
accuracy			0.88	3378
macro avg	0.83	0.71	0.75	3378
weighted avg	0.88	0.88	0.87	3378



Model interpretation

This model's performance with respect to precision and recall of minority class is not a good one. A recall of 0.45 means out of 100 churning customers, model can only identify 45 correctly. The precision is not very high either. So other models may have to be looked at for this exercise. Model coefficients for this model have been provided below:

	Coefficient
Complain_LY	1.916931
Maritalstatus_Single	0.976644
ACSegment_Regular	0.644678
ACSegment_Regularplus	0.574558
City_Tier	0.372012
User_Count	0.323650
CC_Score	0.280958
Gender_Male	0.220045
ACSegment_Superplus	0.209259
Rev_Permonth	0.127603
CC_Contacted_LY	0.032366
Days_Since_CC	-0.069039
Tenure	-0.127919
Payment_Ewallet	-0.158568
Maritalstatus_Married	-0.163270
Logindevice_Mobile	-0.375151
Payment_Debitcard	-0.647071
Payment_UPI	-0.655724
ACSegment_Super	-0.819457
Payment_Creditcard	-0.821820

Observations:

The best model out of all LDA models has given the following coefficients.

Compared to overall best model provided by Gradient boost, we can see that the features are not ranked in the same way. In fact, tenure seems to have a low-ranking coefficient although the model has established the inverse relationship it has with churn.

This is probably why LDA's performance is not comparable to other model's performance.

Complaints made by customers the previous year and Single customers are the top two positive coefficients. If there was a complaint the previous year and if the customer is single, that increases the possibility of churn. Similarly, Credit card payment and Super segment customers have the highest negative coefficients. This implies that customers paying using credit card and customers falling under super segment accounts churn lesser. This model has not performed well compared to other models. Hence the interpretation may not be very useful.

Ensemble method - Gradient Boost

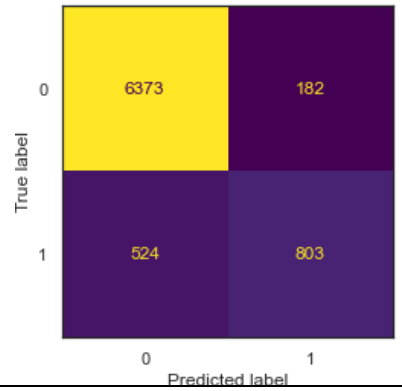
Gradient boost is an ensemble machine learning algorithm that trains underlying models in a gradual, additive and sequential manner. In this modelling exercise, SKlearn's Gradient Boost classifier function was used for modelling:

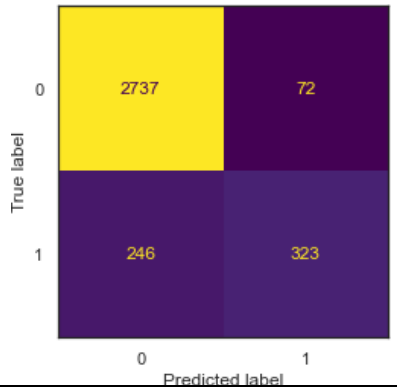
- The base model was run with default hyperparameters and the performance metrics noted. Tuning was later done using GridSearchCV.
- Model performance metrics for base model and best model have been provided in the below sections

The details of this algorithm, its tuning and performance metrics have been provided in this section. The reasons why the tuned Gradient boost was selected as best model and model interpretation are provided :

Gradient boost base model with default hyperparameters

Train dataset Confusion Matrix	Test dataset Confusion Matrix



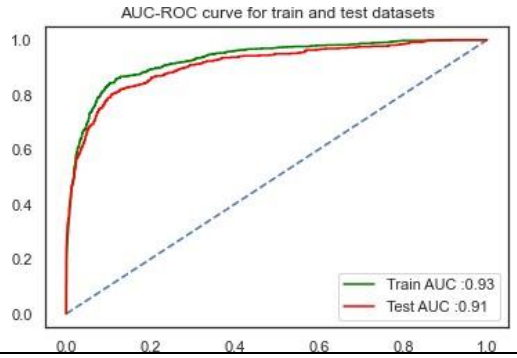


Train dataset Classification report

	precision	recall	f1-score	support
0	0.92	0.97	0.95	6555
1	0.82	0.61	0.69	1327
accuracy			0.91	7882
macro avg	0.87	0.79	0.82	7882
weighted avg	0.91	0.91	0.90	7882

Test dataset Classification report

	precision	recall	f1-score	support
0	0.92	0.97	0.95	2809
1	0.82	0.57	0.67	569
accuracy			0.91	3378
macro avg	0.87	0.77	0.81	3378
weighted avg	0.90	0.91	0.90	3378



The model is robust (no overfit or underfit) but the recall is quite poor in both train and test data. Tuning was done to see if performance can be improved on the model.

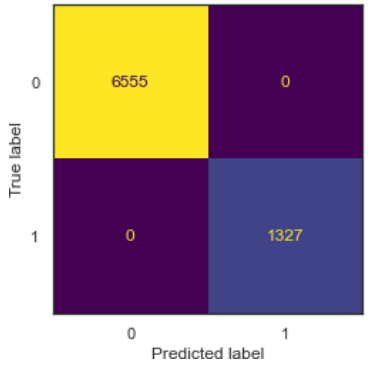
SKLearn GradientBoost - model tuning

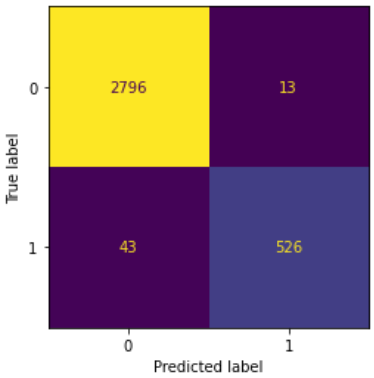
- The model's hyperparameters were tuned using GridSearchCV function and model constructed using the best parameters selected by GridSearchCV.
    - Tuning improved the model performance considerably

Model reference	Data Used			Hyper parameters	Train data					Test data				
	Outliers	Smote	Scaling		Accuracy	Precision	Recall	F1	AUC	Accuracy	Precision	Recall	F1	AUC
GB_model1	Treated	No	No	Default base model	0.91	0.82	0.61	0.69	0.93	0.91	0.82	0.57	0.67	0.91
GB_model2	**Treated**	**No**	**No**	**GridSearchCV, best model for f1 score**	**1**	**1**	**1**	**1**	**1**	**0.99**	**0.99**	**0.94**	**0.96**	**1**

Gradient Boost best model with tuned hyperparameters

Train dataset Confusion Matrix	Test dataset Confusion Matrix





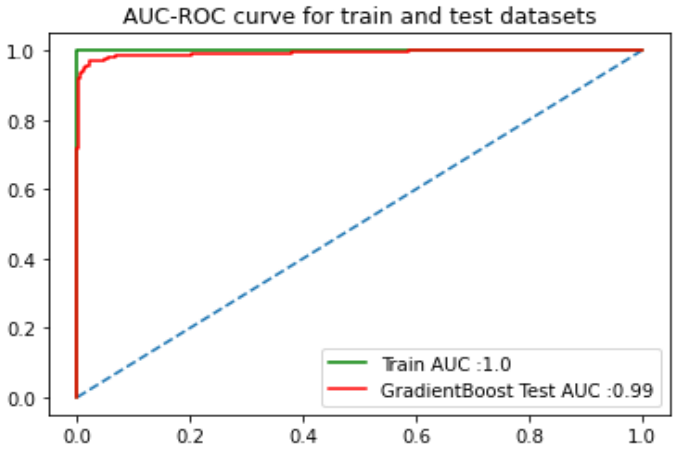
Train dataset Classification report

	precision	recall	f1-score	support
0	1.00	1.00	1.00	6555
1	1.00	1.00	1.00	1327
accuracy			1.00	7882
macro avg	1.00	1.00	1.00	7882
weighted avg	1.00	1.00	1.00	7882

Test dataset Classification report

Accuracy : 0.9834221432800474

	precision	recall	f1-score	support
0	0.98	1.00	0.99	2809
1	0.98	0.92	0.95	569
accuracy			0.98	3378
macro avg	0.98	0.96	0.97	3378
weighted avg	0.98	0.98	0.98	3378



Since the tuned model has shown 1s in all the train data performance parameters, it seems to have learnt all the noise in training dataset well. The test data shows a performance that is within 10% of train data performance. However, in order to ensure that it has not overfitted on train and the test performance can be achieved with unseen data, 5-fold and 10-fold cross validation on entire data was done. The mean f1-score for 5-fold (0.96) was the same as test f1-score (0.96) and 10-fold cross validation showed an increase in mean f1-score (0.98). All the folds also showed individual f1-scores that was comparable or better than test f1-score.

Model validation

- The given data was split into train and test dataset in the ratio 70:30. The test dataset was held out and kept for validation purpose only.
- All models were trained only on the train dataset. The trained model was used to predict train dataset target variable. The performance metrics such as Accuracy, F1-score, Precision, Recall, confusion matrix, ROC curve and AUC was observed and recorded on train dataset.
- The trained model was then used to predict target variable on test dataset. All the above said performance metrics were observed and recorded for the test data performance as well.
- If train data seemed to be giving all 1's and the difference between train and test performances are not off by more than 10%, a validation of test scores was done by doing 5-fold and 10-fold cross validation on entire dataset. The scores on this cross validation were compared to test dataset scores to ensure that model had not overfitted on train dataset.

Criteria for the best performing model

Primary criteria

- **Precision, Recall & F1-score for 1s:** This is the case of class imbalance as the dataset has 16.8% churns. In this case study 'Precision' and 'Recall' of class 1 or the minority class is most important. Combining these 2 metrics, F1-score for class 1 is also used in the comparison.
    - **Precision** is defined as True positive/ (True Positive + False Positive). It answers the question - Out of all customers that the model identifies as churning customers, how many are actual churners? This is the most important factor in this case as budget for offers to retain customers would be limited and more false positives would mean spending that budget on customers who would not churn anyway. The problem statement states that the revenue assurance team is very stringent about providing freebies where it is not required. Translated into metric, this would mean that the precision for 1's/churns should be highest.
    - **Recall** is defined as True Positive / (True Positive + False Negative). It answers the question - Out of all actual churning customers, how many does the model correctly identify as churners This is very important as the purpose of the project is to identify as many churners as possible in order to give special offers in order to retain them. For the DTH provider, customer acquisition cost is very high and hence retention is of utmost importance. This is the whole reason why the project exists and hence Recall for 1s is also important in this case

F1-score is harmonic mean of Precision and Recall. Where both Precision and Recall are important, this metric can be used as a single metric that needs to be optimized for.

Secondary criteria

- **Accuracy:** This is a classification problem and the dataset has class imbalance. That is, the proportion of churn and non-churn customers are not equal. With imbalanced classes, it's easy to get a high accuracy without actually making useful predictions. So, accuracy as an evaluation metric makes sense only if the class labels are uniformly distributed. We are concerned with correct prediction of churn customers (class 1). Hence 'Accuracy' is not a correct metric to compare various models but for the sake of completeness and to ensure that 0s (majority class) are not overlooked, it is still recorded in the comparison matrix.

- **AUROC:** In addition to the above metrics, the Area under curve of ROC curve is also used to evaluate model performance. An ROC curve (or receiver operating characteristic curve) is a plot that summarizes the performance of a binary classification model on the positive class. It is a curve that is constructed by evaluating true positives and false positives for different threshold values. As visualizing ROC curve is difficult for actual comparison, the Area Under Curve (AUC) metric helps with a numeric comparison. The closer the AUC is to 1, the better the model. However, like accuracy this also works well for balanced dataset[4]. For the sake of completeness, this is also recorded in comparison matrix.

		Data Used				Train data					Test Data				
Model reference	Algorithm used	Outliers	Smote	Scaling	Hyperparameters	Accuracy	Precision	Recall	F1	AUC	Accuracy	Precision	Recall	F1	AUC
LR model 1	Logistic Regression	Treated	No	No	Default base model	0.89	0.77	0.51	0.61	0.88	0.89	0.78	0.5	0.61	0.87
LDA model	LDA	Treated	No	No	GridSearch CV, best model for F1 score	0.89	0.77	0.47	0.59	0.88	0.88	0.77	0.45	0.57	0.86
GB model	**Gradient Boost**	**Treated**	**No**	**No**	**GridSearch CV, best model for F1 score**	**1**	**1**	**1**	**1**	**1**	**0.99**	**0.99**	**0.94**	**1**	**1**

A comparison of best model from all algorithm

Why Gradient boost is the best model?

- The hyperparameter tuned Gradient Boost model (GB_model2) has given the best performance in terms of test data precision, recall and f1-score which are the primary evaluation metrics. The Accuracy and AUC are also highest amongst all the models. As the model looked like it overfit on train dataset, a further 5-fold and 10-fold cross-validation was done on complete dataset in which the F1-score on all folds was either comparable or greater than test data f1-score (the test data performance held true or was better for all folds)
- The difference between train data metrics and test data metrics is within 10%
- The model is interpretable. Sklearn provided feature importance for the model

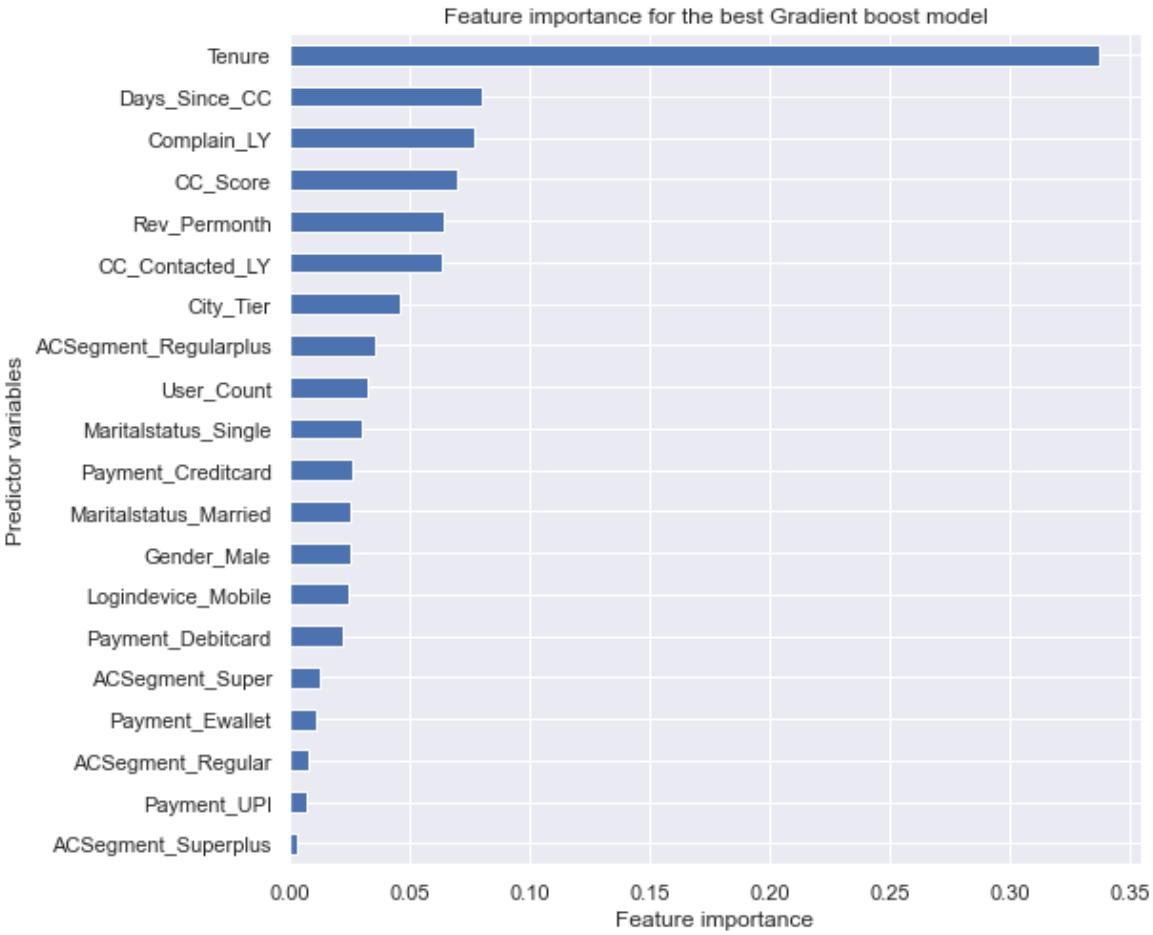
How can business use these metrics?

- A precision of 0.99 implies that out of 100 customers that the model has identified as churned, 99 would actually churn and 1 would not. Any marketing budget allocated for a targeted campaign for retention of these customers would be most optimally utilized as only 1/100 customers would be incorrectly identified as churn.
- A recall of 0.94 implies that for 100 customers who actually churn, the model would have identified 94 as churn and 6 as not-churn. This would mean that the campaign would target these 94 customers and there is scope for retaining these customers and missing out on 6 customers.
- Based on the customer base for which prediction needs to be done, business can use the above to project the numbers of customers who would churn and how many the model would identify and miss.
- That could be further used to come up with per customer budget (if total budget for retention campaign is known) so that appropriate offers can be designed for each customer.
- If per customer budget is known, cost projections for retention campaign can be calculated.
- If limited budget is available and not all customers can be covered, the model can also provide the probability of churning so that high probability customers can be targeted first. Also, a different perspective to this problem would be to do a segmentation and target high value customer segment.

Model interpretation from best model:

Gradient boost model has given very good performance after tuning. The model is robust (no overfit or underfit).

The figures below show the feature importance given by the Gradient Boost model



Top 5 features from Gradient Boost

- From EDA, we can observe that for lower tenures especially within the first month, the churn is higher. Hence, once a customer has been acquired, the first two months is very important to keep the customer satisfied.
- Churned customers had contacted customer care more recently before churning than active customers. Per EDA, median days since last customer connect is higher for active customers. Churned customers had contacted customer care recently before churning.
- The next important parameter to predict customer churn per this model is number of times customer care was contacted by the customer. Per EDA, the median and third quantile of number of times customer care was contacted previous year is higher for churned customers compared to active/current customers.
- According to ranking of important features, except tenure, the other features in top 5 are related to customer care or complaints. This may be indicative of need to monitor and improve the customer care processes.

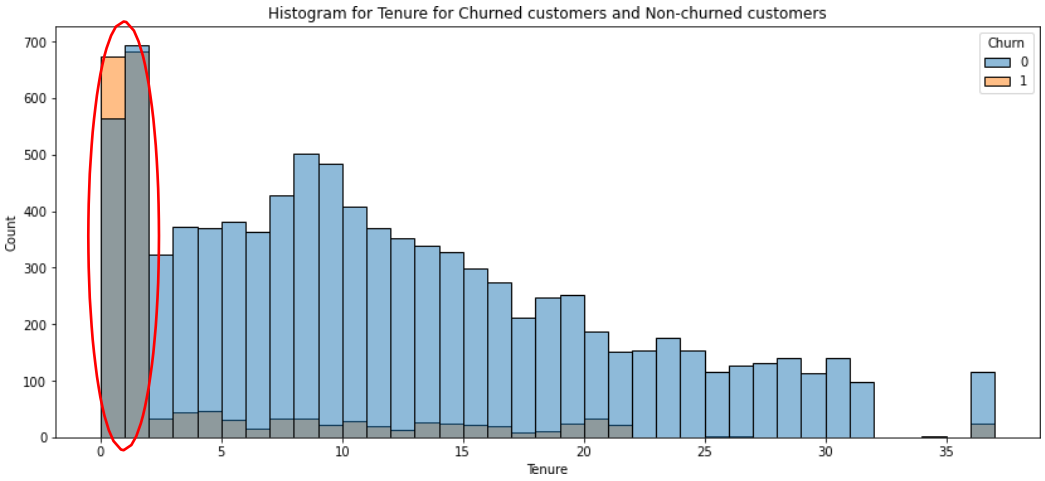
Final interpretation / Recommendations:

Business recommendations:

1.1. Business Insights from EDA

- **Customer feedback:** 78% of customers have rated service as 3 or less than 3 (out of a scale of 5). Likewise, 61% of customers have rated customer care agents a score of 3 or less than 3 (out of a scale of 5). This indicates that the customer feedback is pointing to dissatisfaction or bare satisfaction in service and also in customer care engagement.
- **Relationship between Tenure and Churn:** In the bivariate histogram for Tenure vs Churn, it can be seen that the churn is very high for low tenures. For tenure between 0 and 1, around 51.85% customers have churned. The reason for this churn needs to be drilled down and addressed. Once the tenure increases, the histogram shows that proportion of churn decreases.
- Relationship between Account segment and Churn:
  More customers in Regular_Plus plan seem to churn. A comparison of this plan and competitors plan with same features and for same pricing range could be done to ascertain if the plan or pricing needs to be changed. Also, customer feedback for customers on this plan could be obtained and analyzed to find out why there is more churn in this account segment.
- Relationship between Monthly revenue and Churn:
  The % churn in high revenue customers is slightly more than the churn in lower revenue customers. This is would a cause for concern for the DTH provider that not only is there more churn but more high revenue customers are churning.
- Relationship between Days since customer care connect and Churn
  Days since customer care connect for churned customers is lesser than for active customers. This shows that churn has happened shortly after the customers have contacted customer care.
- Relationship between Customer care contacted last year and Churn
  The number of times customer care was contacted previous year was more in churned customers compared to active customers.
- Relationship between Complaints made last year and Churn
  Proportion of customers who complained is significantly more in churned customers compared to active customers.
- Relationship between User count and Churn
  Proportion of accounts with user counts 5 and 6 is more in churned customers compared to active customers.
- Relationship between Payment type and Churn
  Proportion of customers who have paid through E-wallet and Cash on delivery is more within churned customers compared to active customers.
- Relationship between City tier and Churn
  Proportion of customers who reside in Tier 3 cities is more within churned customers compared to active customers. Whether there is more competition in those cities compared to Tier-1 cities needs to be explored by the DTH provider.
- Relationship between Marital status and Churn
  Single customers have churned more compared to married or divorced customers.

High churn rate in low tenure customers:



Tenure and Churn

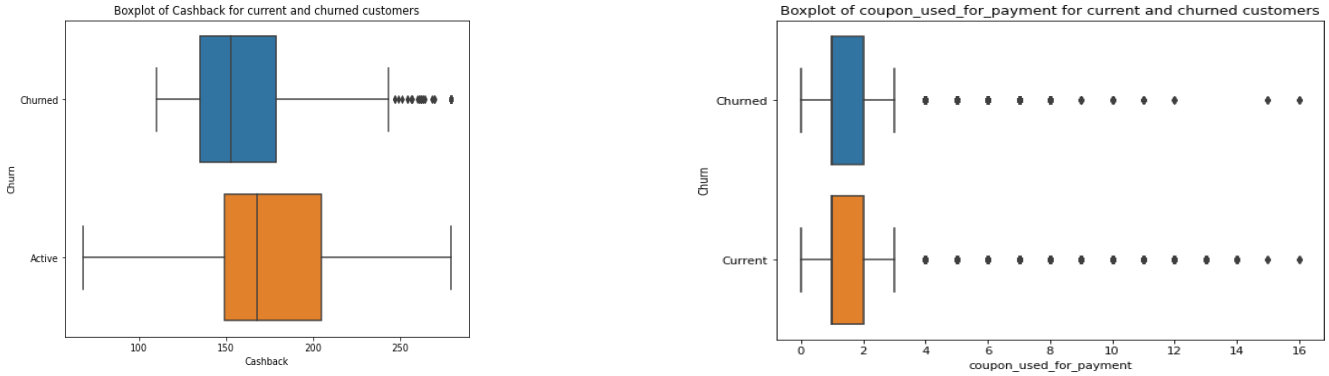
Insight: ***Churn is highest between the tenure period 0 to 2***

Possible Reasons: Bad first experience or Trial periods/prepaid accounts that expire automatically if no top-up is done within a predefined period. Important to determine between the above two reasons. Based on high customer care calls, complaints registered and low cashback and coupons for low tenure customers, it points to the first reason.

Business recommendations:

- **Activation**/Onboarding team could **extend support** beyond the initial setup until customers settle down with the service
- Activation team **proactively engages customers** for the first month or two
- Customer care take a **feedback survey** about the process so that any hiccups can be understood and sorted out
- To increase response rates for feedback, gift cards/coupons can be given.

Existing retention programs – Cashback and Coupons

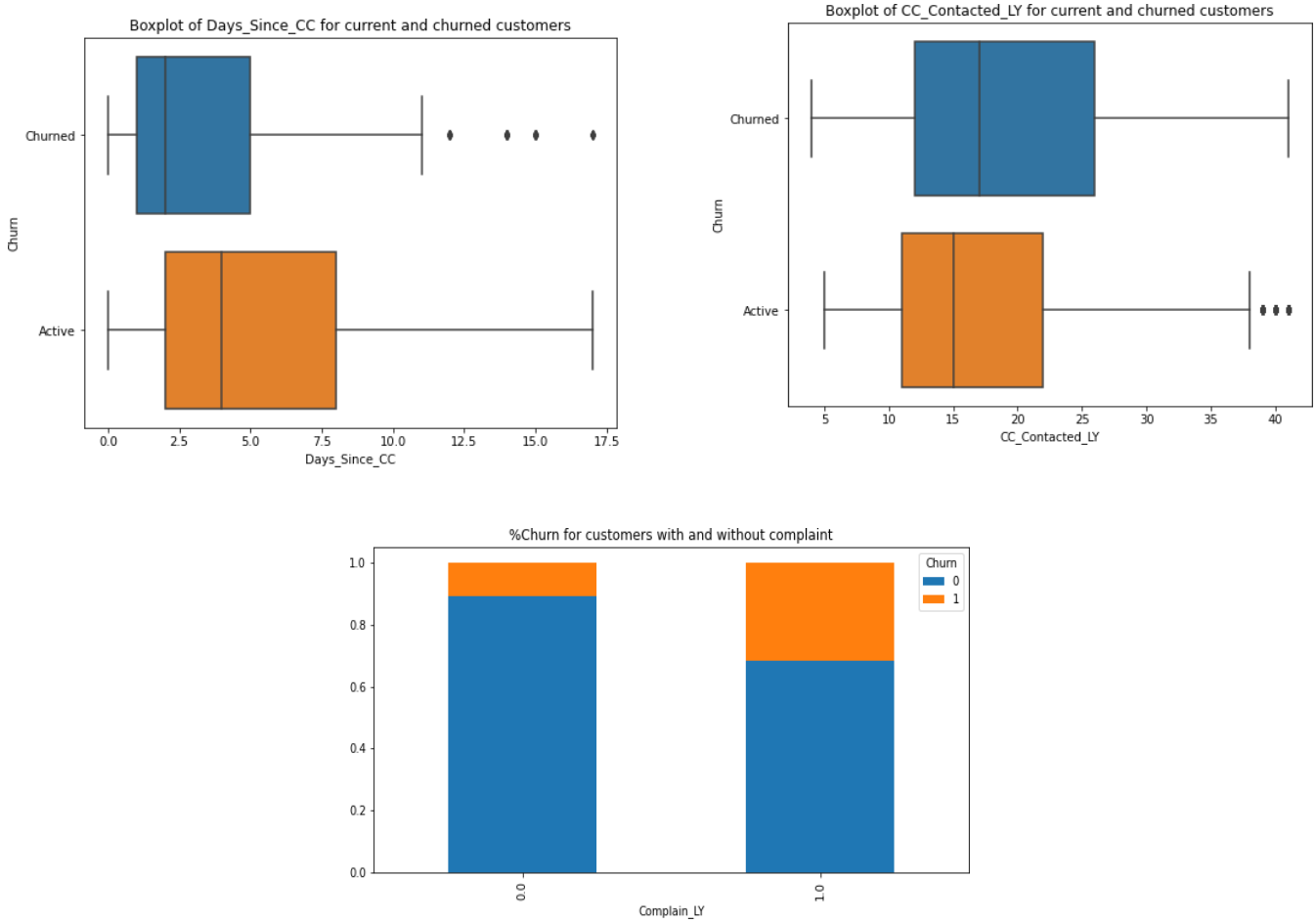


Churn and existing retention

- The churned customers as shown in first boxplot have lesser cashback
- The churned and active customers have almost used the same number of coupons for payment
- *Insight: The current retention programs do not seem to be focusing on the customers with higher risk of churn*

Recommendation: Review whether existing cashback and coupon programs are still relevant given the current churn model. If they are not relevant, design new retention programs to address current high risk customer group.

1.2. Churn and Customer care service



Churn and Customer care service

Churned customers seem to have contacted customer care more recently before churning

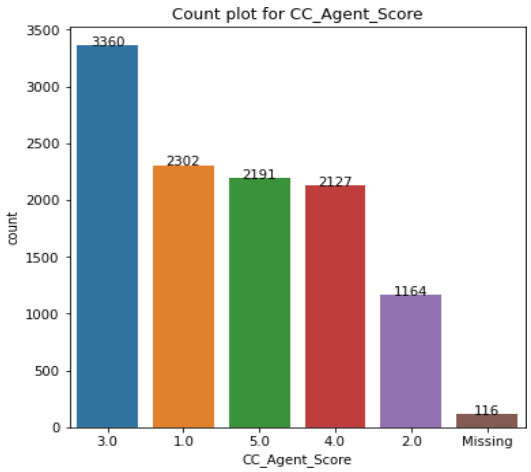
- The number of times churned customers contacted customer care in the year is higher than number of times active customers contacted customer care
- 31% of customers who registered complaint churned Vs 11% of customers who have not registered complaint in last year

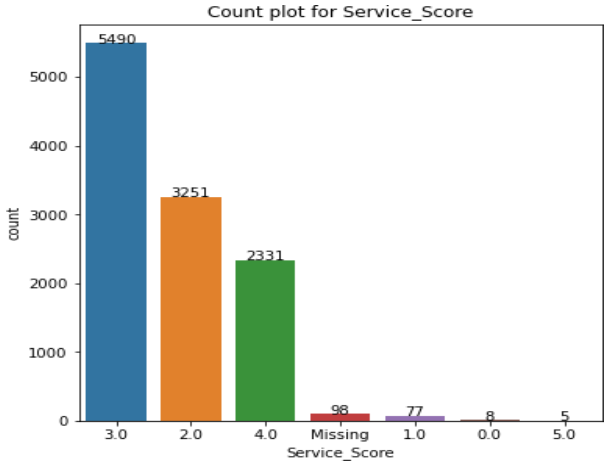
Insight: ***These indicate behavioral changes in customer before churn happens***

Recommendation: Analyze Complaints & Customer care contact reasons

- Perform Root cause analysis, identify and fix top reasons
- Establish Service level agreements (if not already present)

Customer care & Service – Customer perspective



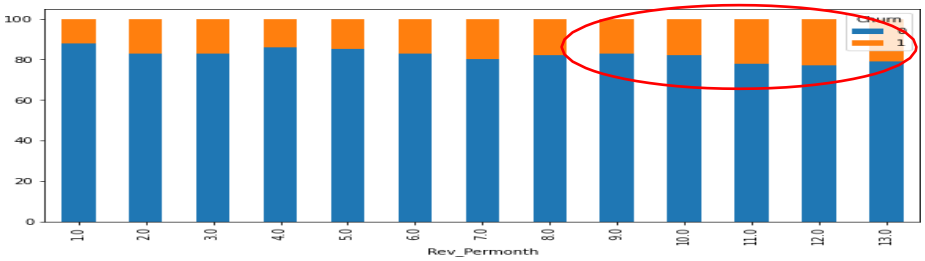


Insight: 78% of customers have rated service as 3 or less than 3

Insight: 61% of customers have rated customer care agents a score of 3 or less than 3

Recommendation: Analyze customer feedback. Perform Sentiment analysis of the feedback (if any) that went along with scores. Identify top reasons that have resulted in low scores; if subjective feedback not captured, capture that as well

Revenue per month and Churn



High churn in high revenue customers

The % churn of customers in higher revenue group, for revenue >=7 per month is higher than low revenue group customers *Insight: More proportion of high revenue customers are leaving compared to less revenue customers*

Recommendation: Create segmented offers for high revenue customers. An illustrative segmentation based on RFM (Recency – Frequency – Monetary) has been shown below along with sample targeted recommendations for each segment. This illustration is based on the test data. A similar segmentation can be done in discussion with client based on what metrics they feel are important to segment based on.

Segmentation basis

Recency (how recently they onboarded): High Recency - Tenure <=2
Frequency (how frequently they have contacted customer care): High Frequency - CC_contacted last year >=17
Monetary (Revenue): High Monetary – Revenue per month >=7

This is just an illustration. The above variables and their thresholds can be changed based on what features business places most emphasis on.

Illustration

	Recency - High		Recency-Low	
	Freq-High	Freq-Low	Freq-High	Freq-Low
Monetary-High	213	195	110	111
Monetary-Low	531	452	137	119

Segment1	629	High Monetary
Segment2	1120	Low Monetary but high recency or frequency
Segment3	119	Low Monetary Low Recency Low Frequency

Segment based recommendations

High monetary Segment 1:
- Cashback /Coupon programs
- Additional free channels
- Periodic Customer feedback Survey

Low monetary-High Recency/Frequency Segment 2:
- Additional free channels
- Activation team support for two months
- Issues to be fixed within SLAs

Low monetary-Low Recency & Low Frequency Segment 3:
- Proactive feedback survey within 2 months
- Issues to be fixed within SLAs

Segmented offers illustration